ARC – ADAPTIVE RETAIL COMPANION: DESIGN AND SIMULATION-BASED EVALUATION OF A RETROFIT AUTONOMOUS SHOPPING CART

Project report submitted
in partial fulfilment of the requirements for the degree of

Bachelor of Technology in Mechatronics and Automation Engineering

by

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DECLARATION

We, Akshada Raghuvanshi (23BMH1110) and Manav Harish Kriplani (23BMH1084), B Tech Mechatronics and Automation Engineering, VIT Chennai, hereby declare that the work being presented in the dissertation entitled "**ARC – ADAPTIVE RETAIL COMPANION: DESIGN AND SIMULATION-BASED EVALUATION OF A RETROFIT AUTONOMOUS SHOPPING CART**" is an authentic record of the work that has been carried out at VIT, Chennai under the guidance of Dr. T. Christo Michael, School of Mechanical Engineering, VIT Chennai.

We hereby declare that the entire work embodied in this dissertation has been carried out by us and no part of it has been submitted for any degree or diploma of any institution previously.

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ABSTRACT

ARC (Adaptive Retail Companion) is a modular kit that retrofits a conventional supermarket cart into an assistive robot for elderly and mobility-impaired shoppers. This project evaluated ARC's ROS 2 navigation stack (RTAB-Map with lidar ICP odometry, a Dijkstra global planner, a TEB-like local controller and a behaviour tree) in a Gazebo store model under a baseline-first protocol. The median localisation error was 2–3 cm. The original stack completed 29 of 30 shopping missions but needed 97 recoveries; all 62 controller aborts came from Nav2's default progress check, which a kinematic calculation shows a slow cart cannot always satisfy after an in-place turn. Ordering the list by path cost with the exact Held–Karp solver cut driving time by 25–41%. Relaxing the progress check to 0.25 m in 20 s together with exact ordering completed all 30 lists in 26% less time and 38% less distance than the original stack (Wilcoxon $p = 5.1\times10^{-7}$). In a bench hardware-in-the-loop test, person detection on a Raspberry Pi 5 drove the behaviour tree and two real drive motors, and the wheels stopped within one 1 s telemetry period of every switch to WAIT. A web shopping interface was connected to the stack through a new ROS 2 bridge. A structural check of the CAD drive mount showed that its wheel bracket must be stiffened. Unmapped obstacles, a goal-reached race and cart-level testing remain open.

Keywords: Assistive robotics, Autonomous shopping cart, ROS 2 navigation, Route optimisation, Hardware-in-the-loop

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LIST OF NOMENCLATURE

v, ω	Linear (m/s) and angular (rad/s) velocity of the robot
v_r, v_l	Right and left wheel linear speeds (m/s)
B	Track width: distance between the driven wheels (m)
D, r_w	Wheel diameter and radius (m)
P_w	Encoder pulses per wheel revolution
s_p	Wheel travel per encoder pulse (mm)
N	Encoder pulses counted in one telemetry period
Δt	Telemetry period (s)
$g(n)$	Least accumulated path cost from the start cell to cell n (m)
$c(p, n)$	Cost of the grid edge from cell p to cell n (m)
$d(P_i, P_j)$	Shortest-path distance between stops i and j (m)
$J(\pi)$	Route cost of visiting order π (m)
$C(S, j)$	Held–Karp cost: least cost from P_0 through set S ending at j (m)
r, T	Progress-check radius (m) and time allowance (s)
$\omega_{max}, \alpha_{max}$	Turn-rate (rad/s) and angular-acceleration (rad/s²) limits
t_{delay}	Sensing, processing and actuation delay (s)
a_b	Braking deceleration (m/s²)
d_{stop}	Stopping distance (m)
m, g	Total moving mass (kg); gravitational acceleration (9.81 m/s²)
C_{rr}	Rolling-resistance coefficient
θ	Floor slope angle (°)
F, τ	Tractive force (N); wheel torque (N·m)
W_{pos}, W_{neg}	Wilcoxon signed-rank sums of positive and negative differences
$\|\dot{\omega}\|$	Mean absolute angular acceleration (rad/s²), a smoothness measure
b, t	Width and thickness of the bracket plate (mm)
Z, I	Section modulus (mm³) and second moment of area (mm⁴)
M, e	Bending moment (N·mm); lateral offset of the wheel load (mm)
F_v	Vertical wheel load on the drive mount (N)
K_t	Stress-concentration factor at the root fillet

c_s, θ_p	Guide clearance per side (mm); resulting rotational play (rad)
R	Turning radius of the drive-axle midpoint (m)
L_{wb}, B_d	Wheelbase and track of the drive wheels (mm)
l_r, h	Centre of mass ahead of the rear axle; height above the floor (mm)
φ_{lat}, a_{tip}	Sideways tipping slope (°); lateral acceleration to tip (m/s²)
l_s, d	Shaft overhang from the bracket face; shaft diameter (mm)
σ_s	Bending stress in the motor shaft (MPa)

LIST OF ACRONYMS AND KEYWORDS

ARC	Adaptive Retail Companion
BT	Behaviour tree
CAD	Computer-aided design
FEA	Finite-element analysis
HIL	Hardware-in-the-loop
ICP	Iterative closest point (lidar scan matching)
Nav2	ROS 2 navigation framework
OSNet	Omni-scale network (person re-identification)
PWM	Pulse-width modulation
RPP	Regulated Pure Pursuit controller
ROS 2	Robot Operating System 2
RTAB-Map	Real-Time Appearance-Based Mapping
SDG	Sustainable Development Goal
SLAM	Simultaneous localisation and mapping
STEP	Standard for the Exchange of Product model data (ISO 10303)
TEB	Timed elastic band
TF	ROS transform tree
UDP	User Datagram Protocol
UI	User interface
YOLO	You Only Look Once (object detector)

KEYWORDS

Assistive robotics, Autonomous shopping cart, ROS 2 navigation, Route optimisation, Hardware-in-the-loop

CHAPTER 1

INTRODUCTION

1.1 Background

Shopping is a routine task that many elderly and mobility-impaired people find tiring: pushing a loaded cart, walking between distant shelves and backtracking through aisles [47]. Service robots that anticipate the needs of older users are an active research area [41], and robotic shopping carts have been demonstrated for visually impaired [50] and elderly shoppers [32]. Most of these systems are purpose-built platforms. A retrofit kit, which adds drive, sensing and computing to the carts a store already owns, is attractive because it keeps the existing fleet and can be serviced as a module [49, 53]. Open-source robotics software now makes such a kit feasible for a student team: ROS 2 [17] and its navigation framework Nav2 [52] provide mapping, planning and control components, and lightweight detectors such as YOLOv8 run on single-board computers.

1.2 The Problem

Joining these components into a reliable assistive robot is harder than choosing each one. A shopping trip is a chain of goals: the item list must be turned into store locations, ordered, driven leg by leg and confirmed on arrival, while the robot keeps its position, avoids shelves and people, and stops when it loses its user. A fault at any interface, such as a default parameter, a missing recovery or a stale message, can stall the trip even when every module works on its own. These faults are rarely measured, because published shopping robots report demonstrations rather than repeated trials. The ARC cart hardware was also not complete during this project, so the navigation software had to be validated before it could be deployed. The problem is therefore to establish, with repeatable measurements, how reliably ARC's integrated software completes shopping trips, which parts limit it, and whether its commands drive real motors safely.

1.3 The Proposed Solution

ARC is a modular retrofit concept: a detachable two-wheel drive module mounted under a standard cart, with a lidar, a camera, an embedded computer and a touch interface. This project (i) designed the cart retrofit and a quick-release drive mount in CAD and checked them structurally; (ii) evaluated the complete ROS 2 navigation stack in a Gazebo model of

a store under a baseline-first protocol, in which the unchanged stack is measured first and every candidate fix is then run on the same tasks and random seeds; (iii) derived engineering models (kinematics, encoder resolution, progress-check timing, stopping distance, drive torque and route-ordering complexity) and checked them against the measurements; (iv) tested the command chain on a bench with real drive motors and on-board person detection; and (v) connected the web shopping interface to the stack. The novelty is not a new algorithm but measured evidence: the faults that limit the system are traced to specific code and configuration, and the recommended fix is shown to complete every test list in about a quarter less time.

CHAPTER 2

REVIEW OF LITERATURE

The review covered four themes that ARC combines: shopping and retrofit robots, mapping and localisation, planning and route ordering, and person perception with behaviour coordination. Sources were peer-reviewed journal and conference papers, with preference for work published from 2019 onwards.

2.1 Assistive Shopping Robots and Retrofit Platforms

RoboCart guided visually impaired shoppers through a grocery store [50]; later carts added ROS-based deep-learning perception [58] and customer-oriented services [46]. Dedicated platforms carry the strongest field evidence: CompaRob assisted elderly users in real grocery stores [32], and Yao et al. tracked shoppers under shelf occlusion in a real supermarket with intersection-aware following [40]. Retrofit carts are older but less developed. Nishimura et al. attached a detachable drive and stereo sensor module to an ordinary cart [53], and Kohtsuka et al. added a driving unit and laser-based following [49]; neither reports map-based navigation of a shopping list. Retrofit autonomy is better established on wheelchairs, for example the modular RTAB-Map-based shared-control system of Ramaraj et al. [28] and detachable propulsion units for manual wheelchairs [11].

2.2 Mapping and Localisation

RTAB-Map is a graph-based SLAM library that supports lidar and visual inputs and long-term operation [13]. Lidar scan matching by ICP gives drift-limited odometry [25], whereas wheel odometry accumulates error without bound [65]. Reviews of localisation [24], visual SLAM [21, 35] and long-term mapping [33] stress re-localisation in changing, repetitive environments such as supermarket aisles. Lidar-inertial odometry [57] and multi-sensor 2D mapping [1] are the main alternatives.

2.3 Planning, Control and Route Ordering

Surveys of mobile-robot navigation separate global planning on a map from local control near obstacles [15, 16, 27, 39]. Dijkstra search returns a least-cost path on a weighted graph [6]. Nav2 provides the controller server, costmaps and behaviour-tree orchestration [18,

52]. Timed-elastic-band control optimises a short trajectory under kinematic limits [31, 55] and has been extended for cluttered spaces [38]; Li et al. fused Dijkstra with an improved TEB and reported a smaller turning radius [14]. Common alternatives are the dynamic window approach [8], its fuzzy variants [54] and Regulated Pure Pursuit [19]. Visiting a shopping list is a sequencing problem related to warehouse order picking [5, 29]. Nearest-neighbour ordering is not optimal in the worst case [30], 2-opt improves a tour without guaranteeing the optimum [4], and the Held–Karp dynamic program is exact for short lists [9]. Closest to this work, Choi et al. ordered supermarket items by Dijkstra distance with a genetic algorithm, in simulation and outside ROS [3].

2.4 Person Perception and Behaviour Coordination

Person-following robots have used stereo detection [56], RGB-D tracking [45, 51] and learned detectors with fuzzy or vision-based control [44, 59]. YOLO detectors are reviewed in [2, 7, 34, 42]; ARC uses YOLOv8n [70] with ByteTrack association [60] and OSNet appearance embeddings [61] for re-identification, a problem surveyed by Wei et al. [36]. Behaviour trees organise such reactive modes [10, 26, 43, 63]. Human-aware and crowd navigation remain open problems [12, 20, 22, 23], and safety requirements for personal care robots are set out in ISO 13482 [64].

2.5 Research Gap

Neither the retrofit cart nor shopping-list ordering is new, so this project claims measured evidence rather than a first. Within the literature reviewed, no study measures, inside a ROS 2 navigation stack, how reliably a retrofit cart completes shopping lists, how far simple ordering rules fall from the optimum, and whether the planned savings survive execution by the local controller. Integration-level faults, which decide whether a trip finishes, are also rarely reported.

2.6 Societal and Industry Significance

Retailers already own large cart fleets. A retrofit module that can be fitted to existing carts and serviced separately would let a store offer assisted shopping without buying new vehicles, and gives older and mobility-impaired customers more independence.

Commercial smart carts such as Caper focus on item recognition and checkout rather than autonomous assistance, and are priced by quotation [67].

2.7 Alternative Technologies and Preliminary Cost-Benefit

Table 2.1 compares the alternatives considered for each subsystem. The chosen options are open source, run on low-cost hardware and can be tested in simulation before the cart is built, which suited a project whose sensors arrived late. A numerical cost comparison with commercial carts is not possible yet because ARC's bill of materials is not final and commercial prices are not public; the cost-benefit evaluation in Chapter 5 is therefore limited to measured benefits and qualitative costs.

Subsystem	Alternatives	ARC choice	Reason
Localisation	Gmapping & AMCL; ORB-SLAM2; LIO-SAM	RTAB-Map & lidar ICP	Open source; lidar and visual inputs; map locking and re-localisation
Global planning	A*; sampling planners	Dijkstra, 8-connected grid	Exact least-cost path; simple and transparent
Local control	DWA; RPP; MPC	TEB-like controller (Nav2)	Handles narrow aisles; 19/20 goals vs 13/20 for RPP
Route ordering	Typed order; nearest-next; genetic algorithm	Exact Held–Karp	Optimal; ≤ 4.8 ms for 12 stops
Person perception	Colour tracking; RGB-D tracking	YOLOv8n & ByteTrack	Runs on a Raspberry Pi 5 with a USB webcam
Mission logic	Finite-state machine	Behaviour tree	Modular reactive modes: LEAD, FOLLOW, WAIT

Table 2.1 : Alternative technologies considered and the reason for ARC's choice

CHAPTER 3

PROBLEM DEFINITION AND OBJECTIVES

3.1 Problem Definition

ARC's navigation software was assembled from separately developed modules, and its end-to-end reliability on shopping trips was unknown. The review shows that shopping robots are usually reported through demonstrations, that shopping-list ordering has not been measured inside a ROS 2 stack, and that integration faults are rarely traced. The cart hardware was also incomplete. The problem addressed is therefore: measure, repeatably and before deployment, how accurately ARC localises, how reliably it completes shopping lists, how its controller and route ordering perform, and how its commands act on real drive motors; identify the causes of failure; and test candidate fixes on the same tasks so that the recommendation rests on paired evidence.

3.2 Objectives

The objectives achieved through this project were:

1. Evaluate ARC's ROS 2 navigation stack in simulation with a baseline-first protocol (localisation, mission reliability, controller, obstacles, odometry, route ordering).
2. Diagnose integration-level faults, derive supporting engineering calculations, and test fixes on the same tasks and seeds.
3. Validate the command chain on a bench with real motors and on-board person detection, and link the shopping interface to the stack.
4. Design a quick-release drive mount for a standard cart, verify its strength and fit, and check the cart footprint against the store aisles.

3.3 Sustainability Goal

The objectives align with the United Nations Sustainable Development Goals [69] as follows. These are intended outcomes; only the route-distance saving has been measured, in simulation.

- **SDG 3 – Good health and well-being:** the cart reduces the physical effort of shopping for elderly and mobility-impaired users.
- **SDG 9 – Industry, innovation and infrastructure:** a retrofit kit upgrades existing carts instead of replacing the fleet.

- **SDG 10 – Reduced inequalities:** assisted shopping supports independent access to stores.
- **SDG 12 – Responsible consumption and production:** reusing carts avoids new vehicles, and optimised routes cut driven distance by 38% per trip in simulation, reducing drive energy.

CHAPTER 4

METHODOLOGY

The project followed the methodology shown in Figure 4.1. Work began with the problem definition and literature review, then split into a mechanical stream and a software stream. The software stream was evaluated in three phases: the unchanged stack was measured first, candidate fixes were then run on identical tasks, and the results were compared pair by pair. Perception, the bench hardware-in-the-loop test and the user-interface link were developed in parallel.

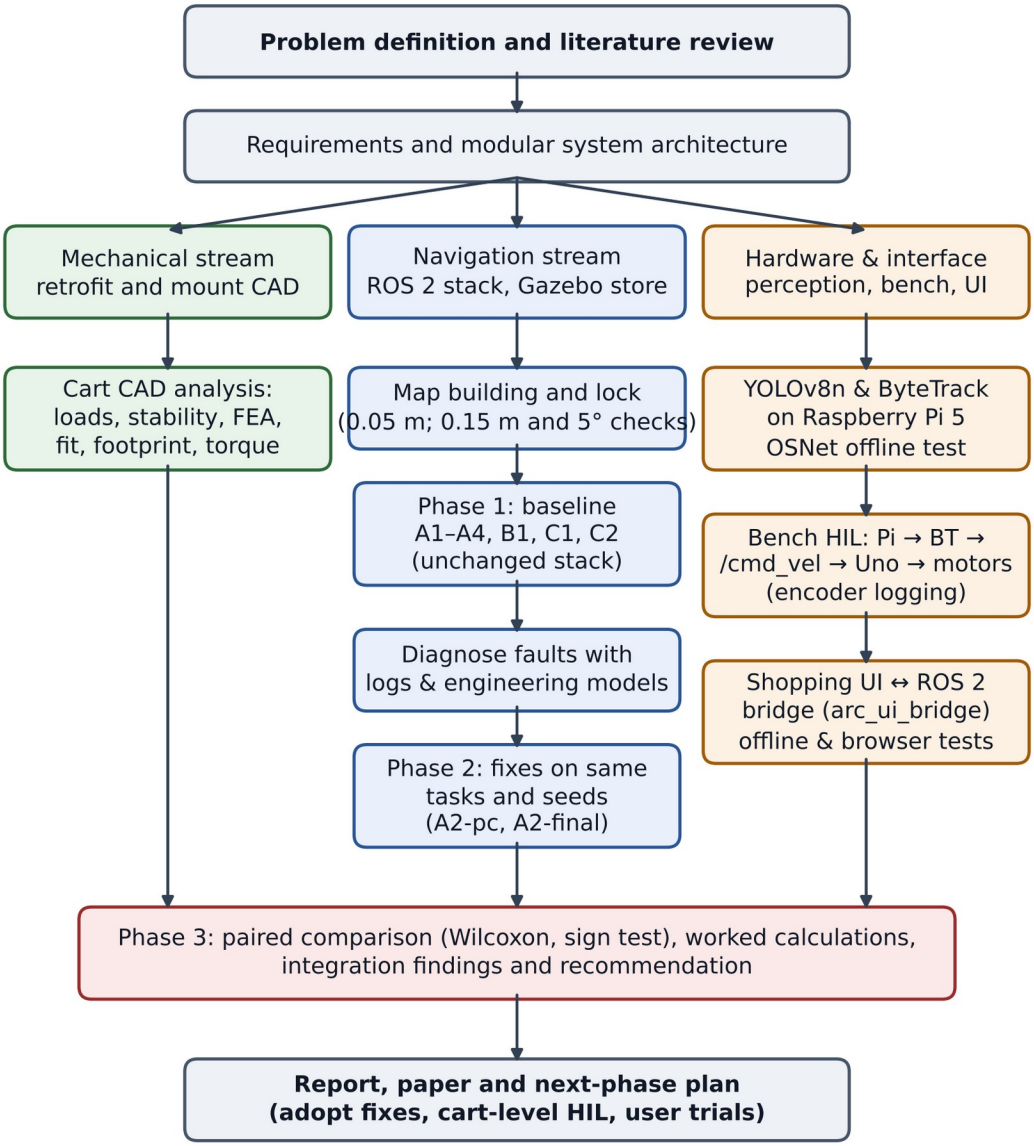


Figure 4.1 : Flow chart of methodology

4.1 Materials and Equipment

The materials and equipment used are listed in Table 4.1. The cart itself was a standard 180 L supermarket trolley (1000 × 610 mm); its retrofit was designed in CAD (Section 4.7). The drive motors were tested on 3D-printed bench mounts.

Item	Specification	Use
Navigation laptop	HP 14s, Ubuntu 24.04, ROS 2 Jazzy, Gazebo Harmonic	Simulation, stack, motor bridge, UI bridge
Embedded computer	Raspberry Pi 5, USB webcam, 5 V / 2.4 A supply	YOLOv8n & ByteTrack person detection
Microcontroller	Arduino Uno, 115 200 baud serial, 300 ms watchdog	PWM to drivers, encoder counting, E-stop input
Motor drivers	2 × SmartElex 15S, PWM limited to ±60 of 255	Drive the two DC motors
Drive motors	2 × DC gear motor, Hall encoder, 7 pulses/rev, 30:1	Bench drive with speed feedback
Bench wheel	Diameter 0.08 m, track 0.16 m	Bench geometry used by the bridge
Simulated robot	TurtleBot3 Burger, 2D lidar, radius 0.105 m	Stand-in for the cart in Gazebo
Software	RTAB-Map, Nav2, Python 3, Ultralytics YOLOv8n	Mapping, control, perception, analysis
CAD and FEA	SolidWorks 2026 (STEP AP214); SimScale linear static	Retrofit design; bracket stress
Geometry and FEA scripts	gmsh 4.15 (OpenCASCADE), scikit-fem 12, Python	Measurements, interference, bracket FEA

Table 4.1 : Equipment and software with specifications

4.2 System Architecture

Figure 4.2 shows the navigation pipeline. A 2D lidar feeds ICP scan-matching odometry [25], and RTAB-Map [13] builds a map and then localises against it. The Dijkstra planner computes the global path on the occupancy grid and replans every 2 s. A hand-over node passes the path to Nav2's controller server, which runs ARC's TEB-like controller. A motion gate scales the velocity command by the behaviour-tree mode (FOLLOW, WAIT, LEAD or emergency stop). The shopping manager sends one goal per item and waits for a 'goal reached' message.

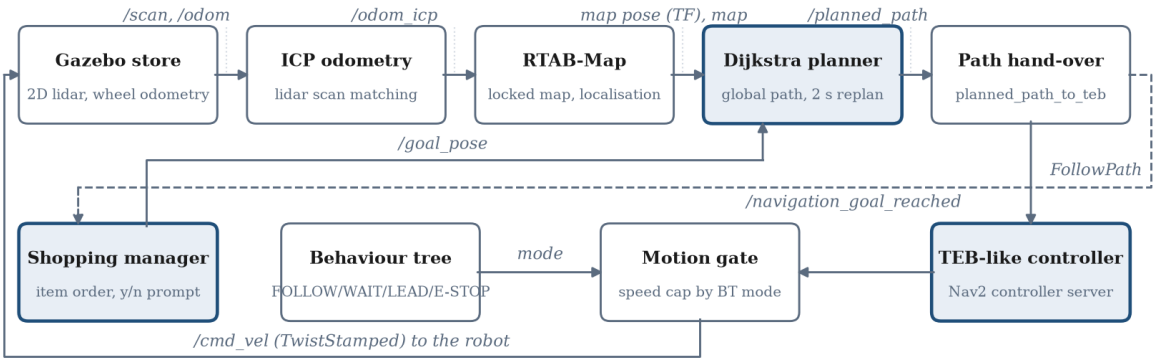


Figure 4.2 : ARC navigation pipeline; shaded boxes are ARC's own planning, control and mission components

Figure 4.3 shows the transform tree recorded with view_frames on the running stack. The chain runs map → icp_odom → odom → base_footprint → base_link, and the two wheel links, the caster, the IMU and the lidar frame (base_scan) hang from base_link. The map → icp_odom transform was published at about 20 Hz, icp_odom → odom at about 5 Hz and odom → base_footprint at about 50 Hz; the wheel links were updated at about 18 Hz, and the caster, IMU and lidar frames are static.

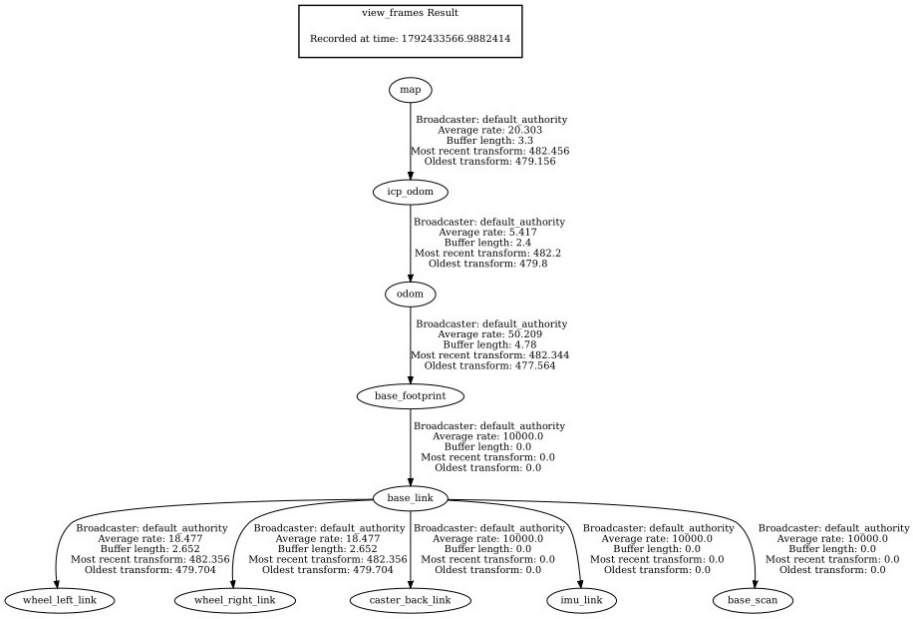


Figure 4.3 : Transform (TF) tree of the running stack, recorded with view_frames

Figure 4.4 shows the node and topic graph recorded with rqt_graph during a simulated run. It confirms the data flow of Figure 4.2: /scan feeds /icp_odometry and RTAB-Map; the Dijkstra planner reads /rtabmap/map, /tf and /shopping_goal and publishes /planned_path; planned_path_to_teb sends the path to the controller server through the follow_path action and publishes /navigation_goal_reached to the shopping manager; the controller's /cmd_vel_nav passes through arc_motion_gate, which publishes /cmd_vel to the Gazebo bridge. On the behaviour side, person_assistant_bt reads the person topics (/person_state, /person_moving, /person_confused, /emergency_stop) and publishes /arc/behavior_mode, /wait_for_person and /emergency_stop_active. In this run the person topics came from a test input node (person_state_input) rather than the camera.

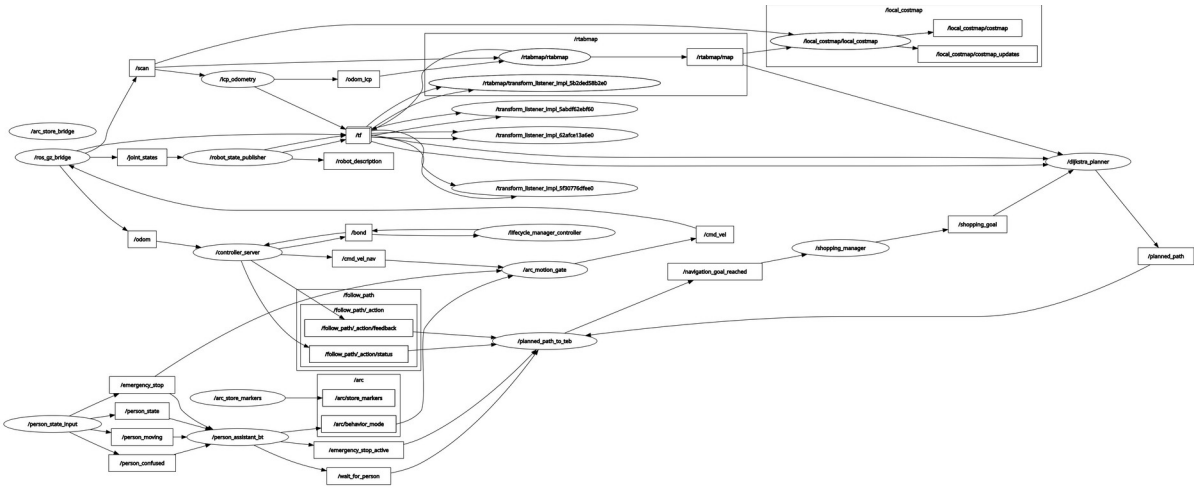


Figure 4.4 : ROS 2 node and topic graph of the running stack, recorded with rqt_graph

4.3 Simulation Environment and Test Protocol

All tests ran in Gazebo [48] in a compact store with three gondola units, a checkout, a tea shelf, a fridge and a produce area: 29 item locations grouped into 9 driving stops. The configuration is given in Table 4.2. ARC's files were not modified, except that the controller's maximum turn rate was lowered from 1.0 to 0.5 rad/s after a fast turn caused an ICP jump in a pilot run. Each session began with an automated 11-waypoint mapping drive, and a map was locked only if the map pose agreed with the true pose within 0.05 m at the start, and then within 0.15 m and 5° during the drive. The tests are listed in Table 4.3. A separate test kit sent goals exactly as the shopping manager does and logged every leg.

Component	Configuration
Map	0.05 m grid, about 180 × 160 cells, rebuilt each session
Localisation	RTAB-Map in localisation mode, lidar ICP odometry
Global planner	Dijkstra, 8-connected, 0.20 m inflation, unknown cells cost ×5, replan every 2 s
Local controller	TEB-like (Nav2): v_{max} = 0.15 m/s, ω_{max} = 0.5 rad/s, α_{max} = 1.0 rad/s²
Behaviour	LEAD mode; motion gate caps speed at 0.10 m/s

Table 4.2 : Simulation test configuration

Test	Task	Trials (seed)
A1	Localisation error, logged in all tests	All samples
A2	Random shopping lists of 3–12 items, typed order	30 (7)
A3	Start–goal pairs, TEB-like controller vs stock RPP	20 each (5)
A4	Pairs ≥ 2.5 m apart, 1–3 boxes placed on the path	20 (9)
B1	First 10 A2 lists with wheel odometry only	10 (7)
C1	Route length by ordering method, offline	50 per size
C2	Lists driven in typed, nearest-next and exact order	30 × 3 (21)
A2-pc	A2 with relaxed progress check (phase 2)	30 (7)
A2-final	A2-pc with exact ordering (phase 2)	30 (7)

Table 4.3 : Navigation evaluation programme

Localisation error was the distance between the map pose and Gazebo's true pose, sampled every 2 s. A leg succeeded when ARC reported 'goal reached' with the robot within 1.0 m of the goal; recoveries counted controller aborts plus goal re-sends (one re-send after an abort or a 40 s stall). Contact was computed from true positions with a 0.105 m robot radius. Paired differences were tested with the Wilcoxon signed-rank test [37], and paired binary outcomes with the sign test.

Figure 4.5 shows the store in Gazebo. Figure 4.6 shows the map RTAB-Map built during the mapping drive: the dotted line is the chain of map nodes along the drive, and the walls, gondolas and produce display appear as closed outlines. Figure 4.7 shows the navigation view used in every test, with the inflated costmap around each fixture, the item locations labelled by aisle and the planned path to the current goal.



Figure 4.5 : The compact store in Gazebo: three gondola units, produce display, tea and coffee shelf, dairy fridge, checkout counter and a shopper

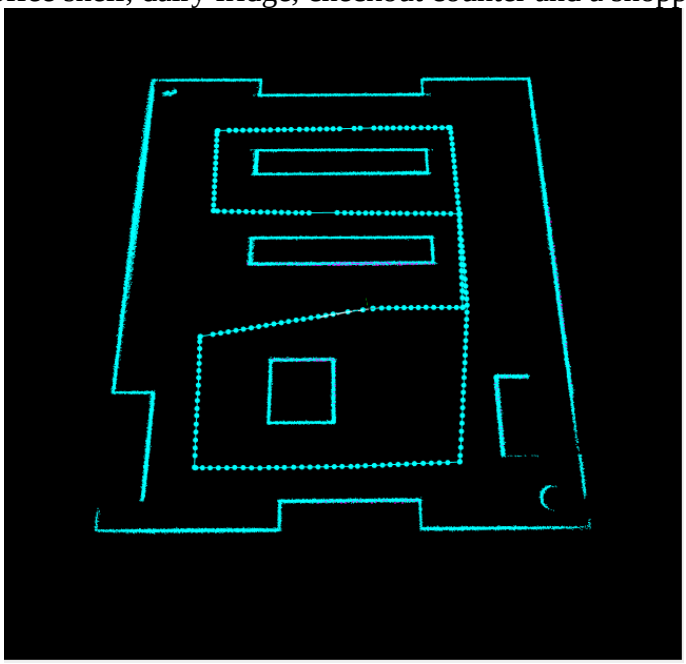


Figure 4.6 : Map built by RTAB-Map during the mapping drive, with the chain of map nodes along the driven route

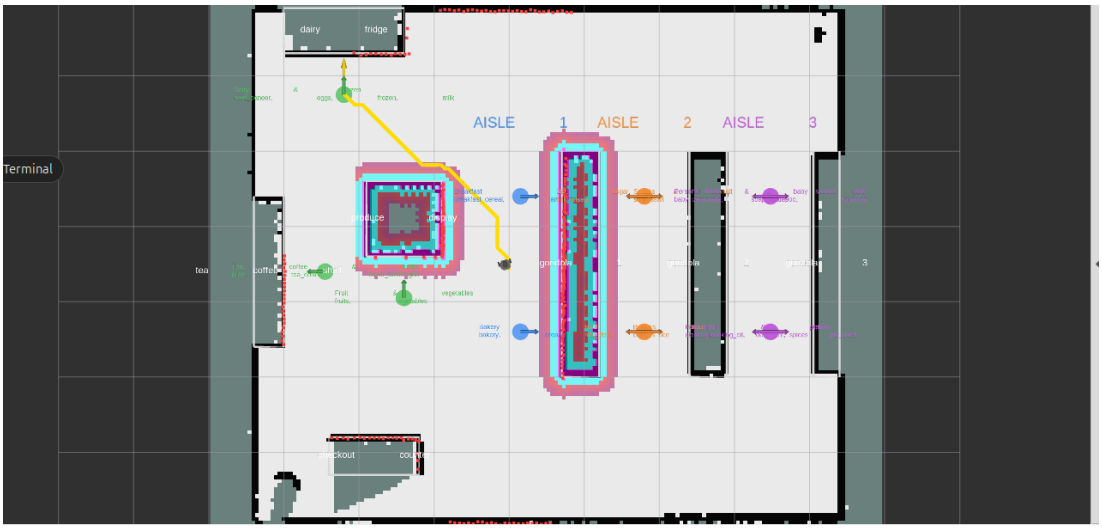


Figure 4.7 : Navigation view in RViz: inflated costmap, item locations by aisle and the planned path (yellow) to the current goal

4.4 Mathematical Models

The following models were used to size the drive, interpret the measurements and choose the fixes. Numerical values are those of the bench hardware or the simulation configuration; assumed values are stated as such.

4.4.1 Differential-drive kinematics

For two driven wheels separated by the track width B, the robot's linear and angular velocities are

$$v = \frac{v_r + v_l}{2}, \omega = \frac{v_r - v_l}{B} \tag{4.1}$$

and the motor bridge inverts them to obtain the wheel targets from the velocity command:

$$v_r = v + \frac{\omega B}{2}, v_l = v - \frac{\omega B}{2} \tag{4.2}$$

With B = 0.16 m, a pure turn command of ω = −0.5 rad/s gives v_r = −0.04 m/s and v_l = +0.04 m/s, i.e. ±40 mm/s at each wheel. The cart-level LEAD speed of 0.10 m/s with no turn gives 100 mm/s at both wheels. The turning radius of the midpoint between the driven wheels is

$$R = \frac{v}{\omega} = \frac{B}{2} \cdot \frac{v_r + v_l}{v_r - v_l} \tag{4.3}$$

so v = 0.10 m/s with ω = 0.5 rad/s gives R = 0.20 m; equal and opposite wheel speeds give R = 0, a turn on the spot.

4.4.2 Encoder resolution and speed measurement

Each motor's Hall encoder gives 7 pulses per motor revolution, and the 30:1 gearbox gives P_w = 210 pulses per wheel revolution. The wheel travel per pulse and the measured speed are

$$s_p = \frac{\pi D}{P_w} = \frac{\pi \times 80\,mm}{210} = 1.197\,mm,\; v_{meas} = \frac{N s_p}{\Delta t} \tag{4.4}$$

With Δt = 1 s, the speed resolution is one pulse, 1.2 mm/s, which is 1.2% of the 100 mm/s LEAD speed. For example, N = 86.5 pulses in one second corresponds to 86.5 × 1.197 = 103.5 mm/s.

4.4.3 Global path planning

Dijkstra's algorithm [6] keeps, for every grid cell n, the least accumulated cost from the start:

$$g(n) = \min_{p \in N(n)} \left[g(p) + c(p,n) \right], c(p,n) = \begin{cases} 0.05\,m & \text{straight} \\ 0.05\sqrt{2} = 0.0707\,m & \text{diagonal} \end{cases} \tag{4.5}$$

The 180 × 160 grid has V = 28 800 cells and at most E = 8V = 230 400 edges. With a binary heap the search costs O((V + E) log V) ≈ 259 200 × 14.8 ≈ 3.8 × 10^6 elementary steps, small enough to replan every 2 s on a laptop.

4.4.4 Route cost and exact ordering

For a visiting order π of n stops starting at the cart's position P_0, the route cost is

$$J(\pi) = \sum_{k=0}^{n-1} d\left(P_{\pi(k)}, P_{\pi(k+1)}\right) \tag{4.6}$$

where d is the Dijkstra path length between stops. The Held–Karp recursion [9] finds the order that minimises J exactly, where C(S, j) is the least cost of a route from P_0 through every stop in S ending at j:

$$C(S,j)=\min_{i\in S,i\neq j}\left[C(S-\{j\},i)+d(P_i,P_j)\right],C(\{j\},j)=d(P_0,P_j) \tag{4.7}$$

Its cost is $O(n^2 2^n)$. For the largest list, n = 12, this is $12^2 \times 2^{12} = 589\,824$ operations over $12 \times 2^{12} = 49\,152$ states, against 12! = 479 001 600 orders for brute force, about 800 times fewer. The measured solve time was at most 4.8 ms.

4.4.5 Progress-check model

Nav2's progress checker aborts a leg unless the robot moves a radius r away from its last reference position within a time T; the default is r = 0.5 m, T = 10 s. The mean speed it demands is r/T = 0.05 m/s, half of the 0.10 m/s LEAD speed. An in-place turn through an angle $\Delta\psi$ moves the robot centre almost nothing. With turn rate ω_{max} and angular acceleration α_{max}, the turn takes

$$t_{turn}=\frac{\omega_{max}}{\alpha_{max}}+\frac{\Delta\psi}{\omega_{max}} \tag{4.8}$$

and the displacement still possible within the check window is

$$\Delta s=v(T-t_{turn}) \tag{4.9}$$

The fixes were judged against this condition, which requires $\Delta s \geq r$. Worked calculations follow in Section 5.2.

4.4.6 Stopping distance

The stopping distance after a stop request is the travel during the delay plus the braking distance:

$$d_{stop}=v\,t_{delay}+\frac{v^2}{2a_b} \tag{4.10}$$

4.4.7 Drive force, torque and power

For a cart of mass m accelerating at a on a slope θ with rolling-resistance coefficient C_{rr}, the tractive force, the torque at each of the two driven wheels of radius r_w, and the wheel power are

$$F = m\left(a + g\,C_{rr}\cos\theta + g\sin\theta\right),\ \tau_w = \frac{F\,r_w}{2},\ P_w = \frac{F\,v}{2} \tag{4.11}$$

4.4.8 Footprint check

The cart's rectangular footprint (length L = 1.00 m, width W = 0.61 m) was bounded by two circles for driving straight and for turning on the spot, with a clearance margin c = 0.05 m:

$$R_{straight} = \frac{W}{2} + c = 0.355\,m,\ R_{turn} = \frac{\sqrt{L^2 + W^2}}{2} = 0.586\,m \tag{4.12}$$

4.4.9 Statistical tests

For paired binary outcomes, the one-sided sign test gives the probability p of k or more 'wins' in n discordant pairs under the null hypothesis of no difference:

$$p = \sum_{i=k}^{n} \binom{n}{i} 0.5^n \tag{4.13}$$

For paired continuous measures, the Wilcoxon signed-rank statistic W_{pos} is the sum of the ranks R_i of the absolute differences that come from positive differences d_i; its mean and standard deviation under the null hypothesis, and the normal-approximation score z, are

$$W_{pos} = \sum_{d_i > 0} R_i,\ \mu_W = \frac{n(n+1)}{4},\ \sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{24}},\ z = \frac{W_{pos} - \mu_W}{\sigma_W} \tag{4.14}$$

4.5 Shopping Interface and UI–ROS Bridge

The browser interface offers a 60-product catalogue, quantities, a shopping list and a route page. A new ROS 2 package (arc_ui_bridge) serves the interface from the laptop and connects it to the stack without modifying any ARC package. It maps the 60 products to 22 of the 29 store items, merges products that share a spot into 9 or fewer stops, orders them by Dijkstra path cost on the live map with the Held–Karp solver, and sends them one at a time on the same topics as the shopping manager. Because 'goal reached' is a latched message, an arrival is accepted only if

$$\left\| p_{robot} - p_{goal} \right\| \leq 1.0\,m \text{ and } t - t_{goal} \geq 3\,s \tag{4.15}$$

where 1.0 m is the planner's goal-snapping radius and 3 s exceeds the planner's 2 s replan period, so a re-sent arrival from the previous goal is never taken. Pause and stop hold the robot at its current pose and cancel the controller's path.

4.6 Perception and Hardware-in-the-Loop Method

Person detection used YOLOv8n with ByteTrack on the Raspberry Pi 5 at a 320-pixel input. Because Debian 13 on the Pi has no ROS 2 packages, the Pi sent person-state flags to the laptop over UDP; both were on a phone hotspot and the Pi was operated over SSH from the laptop. A relay republished the flags for the behaviour tree and held it in WAIT if no data arrived for 1.5 s. A motor bridge converted the velocity command into wheel targets with Eq. (4.2), mapped them to PWM through a per-wheel calibration and logged every encoder report. The Uno firmware limited PWM to ±60 of 255, stopped the motors if no command arrived for 300 ms and refused to drive while the E-stop input was open. The rest of the stack ran in simulation, so the real motors and the simulated robot received the same commands. OSNet re-identification was tested offline on a 31 s recording (465 frames at 15 fps).

4.7 Mechanical Design and Structural Analysis

The retrofit was modelled in SolidWorks 2026 (Figure 4.8). The host cart's basket, frame and handle are unchanged. Two drive modules replace the front wheels and form a differential drive; the two rear wheels remain passive. Each module consists of a quick-

release mount, a geared DC motor and a 150 mm wheel. The motor is bolted to the outer face of the mount's wheel bracket, and its 15 mm output shaft passes through the bracket to carry the wheel on the inner side. The overall dimensions of the retrofitted cart are given in Table 4.4.



Figure 4.8 : CAD assembly of the ARC retrofit cart with the two drive modules at the front wheels

Quantity	Value	Quantity	Value
Overall length	904 mm	Wheel diameter	150 mm
Overall width (motors)	667 mm	Drive / rear wheel width	30 / 33 mm
Cart body width	440 mm	Wheelbase	680 mm
Overall height	1 275 mm	Track of the drive wheels	340 mm
Lowest part above floor	42.5 mm (bracket)	Wheel offset from bracket	71.7 mm

Table 4.4 : Principal dimensions of the retrofitted cart, measured from the CAD assembly

The quick-release mount has four parts (Figure 4.9, Table 4.5): a sleeve fixed to the cart through an adaptor plate, made of a top plate and a 50 mm cylinder with two guide slots 15.2 mm wide; a slider, made of a 70 × 70 mm plate and two arc-section guide rails 13 mm wide that run up into the slots; a wheel bracket, a 3.5 mm plate with an 18 mm hole for the motor boss, two 4.2 mm motor-screw holes and 7.5 mm root fillets; and two flat springs that latch the rails at the top plate. To fit the module, the rails are slid up into the sleeve until the springs engage; to remove it, the springs are compressed and the slider is drawn out. The guide rails and sleeve carry the transverse loads and moments, and the wheel's vertical load passes from the bracket into the slider plate, which bears on the base of the sleeve. The springs only retain the module along the removal axis.

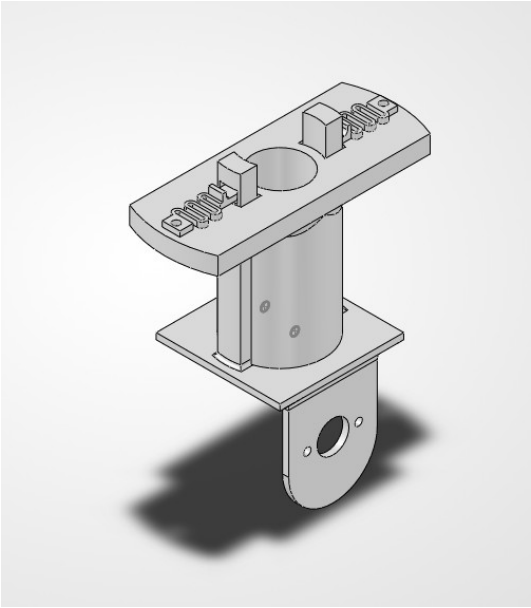


Figure 4.9 : Quick-release drive mount: sleeve with the two spring latches (top), slider plate and wheel bracket (bottom)

Part	Function	Volume (mm^3)	Mass Al / steel (g)
Sleeve	Fixed to cart; guide slots	129 224	349 / 1014
Slider	Plate and two guide rails	39 145	106 / 307
Wheel bracket	Carries motor and wheel	10 761	29 / 85
Flat spring (×2)	Latch	818 each	2.2 / 6.4 each
Total	Mount without motor and wheel	180 766	488 / 1419

Table 4.5 : Parts of one quick-release drive mount, from the STEP files

The STEP files of the 10 parts (21 instances) and of the two assembly levels were read in code (gmsh with OpenCASCADE). Each part was placed with the transforms stored in the assemblies, and the placed geometry was used to measure the dimensions, wheel positions and offsets, compute volumes and masses (density 2.70 g/cm^3 for aluminium and 7.85 g/cm^3 for steel), check every pair of parts in a drive module for interference, and measure the fit of the rails in the slots. The measured geometry then fed four analyses: the static wheel loads and tipping limits of the loaded cart; the swept envelope of the cart, which was re-run through the store-map footprint check (Section 4.4.8); the drive torque; and the strength of the wheel bracket.

The bracket was analysed twice. The team's SimScale model of the slider and bracket was a linear static analysis with second-order tetrahedral elements (6 689 elements, 12 446 nodes), the bracket-to-plate interface fixed and a load at the motor hole. Its material and load were recovered from the exported result fields: the ratio of stress to strain gives E = 70 GPa and $\nu = 0.34$, i.e. aluminium, and integrating the internal forces over plate cross-sections gives about 10 N normal to the plate and 10 N in its plane. A second, independent model was then built in Python with scikit-fem: the bracket from its STEP file, meshed with quadratic tetrahedra (96 231 degrees of freedom, and 167 157 for a mesh-convergence check), the base fixed, E = 70 GPa and $\nu = 0.33$. It was first run with the SimScale load to check it against SimScale, and then with the design wheel load, applied as the vertical force on the motor hole together with the moment of its 71.7 mm offset, carried into the plate as a

linearly varying pressure under the motor flange. Both models were checked against beam theory using the models below.

4.7.1 Bracket bending

The bracket plate (width b = 50 mm, thickness t = 3.5 mm) was treated as a cantilever from its root. Its section modulus, second moment of area, bending stress and the deflection at a distance L from the root under a force F applied at a distance a are

$$Z=\frac{b t^2}{6}=102.1\,mm^3, I=\frac{b t^3}{12}=178.6\,mm^4, \sigma=\frac{M}{Z}, \delta=\frac{F a^2(3L-a)}{6EI} \tag{4.16}$$

Because the wheel sits beside the bracket on the motor shaft, its vertical load F_v acts at a lateral offset e from the plate and bends it with a moment that is the same along the whole plate. The peak stress at the root fillet follows from the stress-concentration factor K_t found by the FEA, and the wheel tilt from the plate rotation:

$$M=F_v e, \sigma_{peak}=K_t\frac{F_v e}{Z}, \theta=\frac{M L_h}{EI} \tag{4.17}$$

where L_h = 32.5 mm is the distance from the plate root to the shaft centre.

4.7.2 Guide clearance and alignment

With slot width w_{slot} and rail width w_{rail}, the clearance per side, and the rotation of the module about the sleeve axis that this clearance allows at the mean rail radius r_m, are

$$c_s=\frac{w_{slot}-w_{rail}}{2}, \theta_p\approx\frac{c_s}{r_m} \tag{4.18}$$

4.7.3 Spring latch

Each flat spring was approximated as a cantilever strip of width w, thickness t, free length L_s and modulus E. The force and bending stress at an end deflection δ are

$$F\approx\frac{E w t^3 \delta}{4L_s^3}, \sigma\approx\frac{3Et\delta}{2L_s^2} \tag{4.19}$$

4.7.4 Wheel loads, tipping and swept envelope

With the centre of mass a distance l_r ahead of the rear axle on a wheelbase L_{wb}, at a height h above the floor and midway between wheels on a track B_d, the static load on each drive wheel, the tipping angles and the lateral acceleration at which the cart would tip are

$$F_v = \frac{m\,g}{2}\cdot\frac{l_r}{L_{wb}},\quad \tan\varphi_{lat} = \frac{B_d/2}{h},\quad a_{tip} = g\,\frac{B_d/2}{h} \tag{4.20}$$

When the cart turns on the spot about the midpoint of the drive axle, a point of the footprint at plan coordinates (x, z) sweeps a circle of radius

$$r = \sqrt{x^2 + z^2} \tag{4.21}$$

4.7.5 Motor shaft bending

The wheel load acts on the 15 mm motor shaft at a lever arm l_s from the bracket face; the bending stress in a solid round shaft of diameter d is

$$\sigma_s = \frac{32\,F_v\,l_s}{\pi\,d^3} \tag{4.22}$$

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Localisation Accuracy (A1)

With ICP odometry the median localisation error was 0.018–0.038 m in every configuration (Figure 5.1), and 91.2–97.9% of samples were inside the 0.15 m and 5° band. Excursions were mostly in heading (median 0.5–1.0°). All large errors came from two events, not gradual drift: lidar odometry failed after contact with boxes in A4 and once with a shelf in A2-pc.

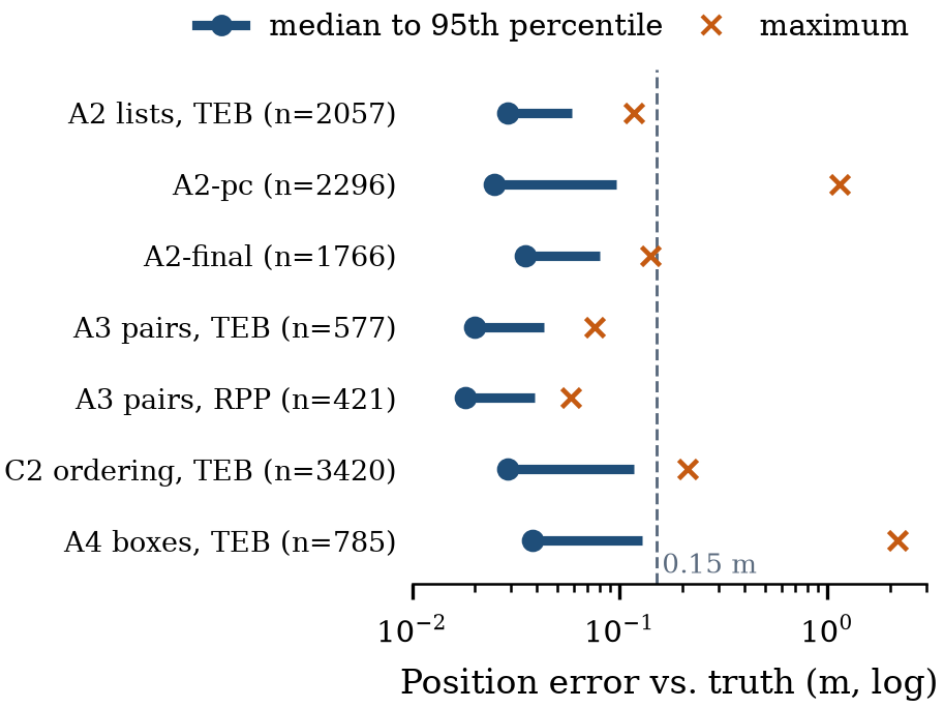


Figure 5.1 : Localisation error per configuration with ICP odometry (samples every 2 s)

5.2 Mission Reliability and the Progress Check (A2)

The original stack completed 29 of 30 missions and reached 143 of 144 stops without contact, but only 96 stops (67%) at the first attempt, and it needed 97 recoveries. All 62 controller aborts were Nav2 code 105, 'failed to make progress'. ARC defines no progress checker, so Nav2's default applies. Applying Eq. (4.8) with ω_{max} = 0.5 rad/s and α_{max} = 1.0 rad/s² explains the aborts:

- A 90° turn takes 0.5 + 1.571/0.5 = 3.64 s, leaving $\Delta s = 0.10 \times (10 - 3.64) = 0.64$ m ≥ 0.5 m: it passes, but with only 0.14 m to spare, which slowing near an aisle entrance or a goal easily consumes.
- A 180° turn takes 0.5 + 3.142/0.5 = 6.78 s, leaving $\Delta s = 0.10 \times (10 - 6.78) = 0.32$ m < 0.5 m: an abort is certain even at full LEAD speed.
- With the relaxed check (r = 0.25 m, T = 20 s) the same turn leaves $\Delta s = 0.10 \times (20 - 6.78) = 1.32$ m, more than five times the radius; the demanded mean speed falls from 0.05 to 0.0125 m/s.

Figure 5.2 shows these budgets. Because the path hand-over has no recovery, each abort would leave the shopping manager waiting indefinitely without the test kit's re-send.

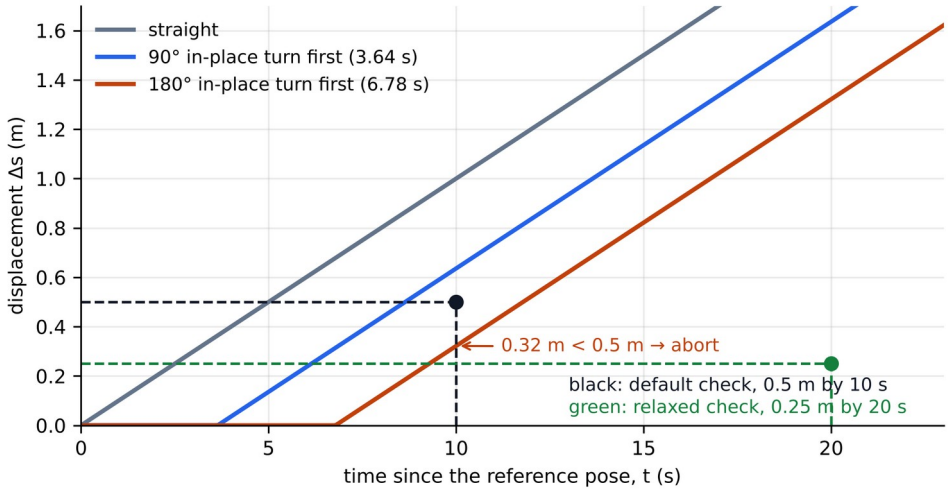
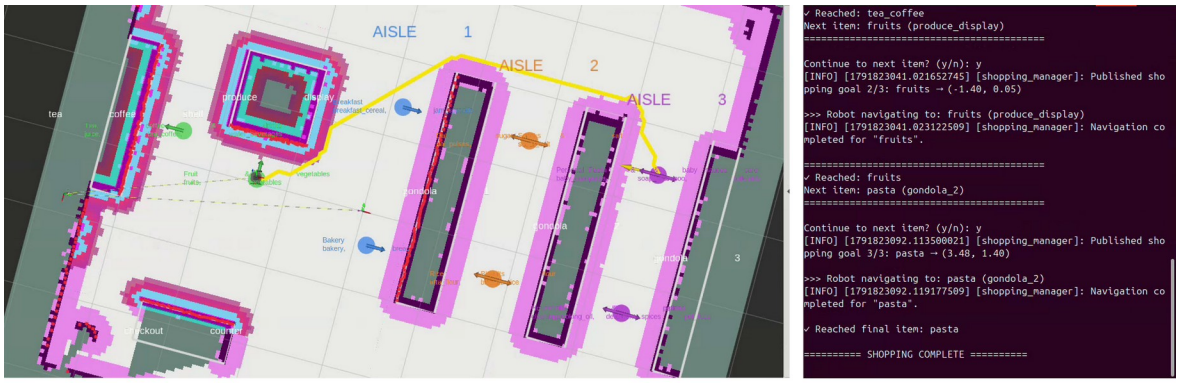


Figure 5.2 : Displacement achievable after an in-place turn at 0.10 m/s against the default and relaxed progress checks

Before the timed tests, the full chain was run interactively with a three-item typed list (tea_coffee, fruits, pasta). The shopping manager published one goal per item, the robot

drove from the produce display through the aisles to gondola 2, and all three stops were reached (Figure 5.3).



(a) (b)

Figure 5.3 : Interactive shopping run: (a) planned path from the produce display to gondola 2 in RViz; (b) shopping manager log, from the second goal to 'shopping complete'

5.3 Local Controller Comparison (A3)

Table 5.1 compares the controllers on the same 20 pairs, with Pure Pursuit limited to the same speed, turn rate and angular acceleration. Pure Pursuit mostly failed by refusing to move, because its collision check found no admissible command in aisles narrowed by the 0.20 m inflation (code 106). The TEB-like controller was more reliable but less smooth: mean $|\dot{\omega}|$ was 0.220 against 0.083 rad/s² (Wilcoxon, $n = 19$, $p = 2.7 \times 10^{-5}$). On the 12 pairs both completed, time (68.2 vs 62.8 s) and distance (4.73 vs 4.29 m) did not differ significantly ($p = 0.23$ and 0.38).

Measure	TEB-like	RPP (stock)
Goals reached	19/20	13/20
Reached at first attempt	12/20	10/20
Recoveries	17	24
Dominant abort	105 (no progress)	106 (no valid control)
Mean $\|\dot{\omega}\|$ (rad/s²)	0.220	0.083
Steering reversals per m	1.88	0.80
Median localisation error (m)	0.020	0.018

Table 5.1 : Controller comparison on 20 identical start–goal pairs (A3)

5.4 Unmapped Obstacles (A4)

Boxes of 0.20 × 0.20 × 0.35 m were placed on the planned path after planning. The stack reached 3 of the 18 goals with boxes and touched a box in 17 (Table 5.2); the median minimum clearance was −0.027 m. The cause is architectural: the planner uses the locked map and never sees the boxes, and the hand-over ignores new paths while one is being followed, so avoidance is left to the local controller alone. It can bend around one box but not pass two or three. In two 3-box legs ICP odometry failed after contact and never recovered.

Boxes	Legs	Goal reached	Contact
1	7	3	6
2	4	0	4
3	7	0	7
All	18	3 (17%)	17 (94%)

Table 5.2 : Outcome by number of boxes on the path (A4)

5.5 Wheel-only Odometry (B1)

With wheel odometry in place of ICP, localisation was as accurate as before while it held (median 0.023 m), but it was lost twice, after about 73 m and 48 m of driving (Figure 5.4). RTAB-Map's correction absorbed 1–2 m of wheel drift until it froze; its log shows correct matches rejected as 'wrong loop closure' because they disagreed with the drifted odometry. Lidar odometry therefore protects re-localisation.

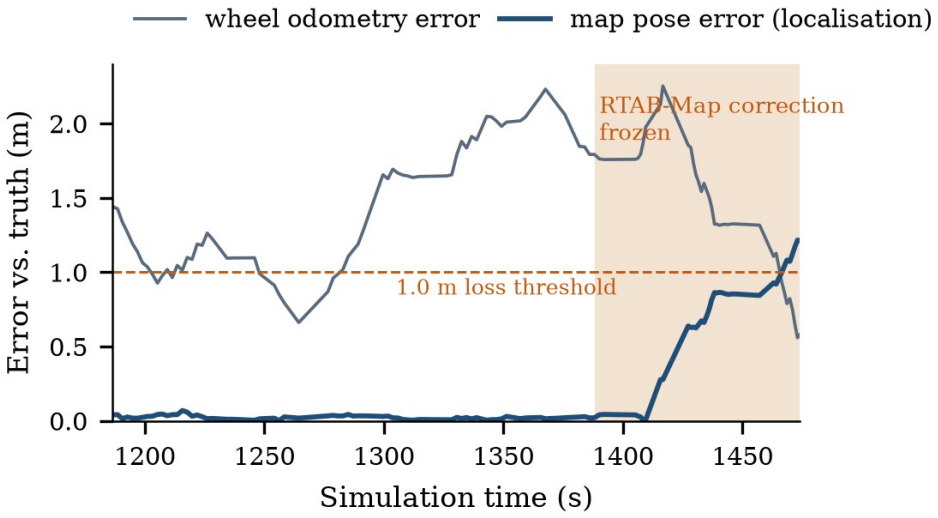


Figure 5.4 : First localisation loss with wheel-only odometry (B1)

5.6 Shopping-List Ordering (C1, C2)

Offline, typed order produced routes 23–94% longer than the optimum for 3 to 12 items (Figure 5.5). Nearest-next on path cost stayed within 1–7%, better than nearest-next on straight-line distance (3–9%), because shelves block the direct line. Driven by the full stack, the exact order was faster than typed order on all 30 lists (Figure 5.6). By the sign test (Eq. (4.13)) with $k = n = 30$, $p = 0.5^{30} = 9.3 \times 10^{-10}$. Median route length fell by 32%, 42% and 51%, and time by 25%, 34% and 41%, for 5, 8 and 12 items. Completion did not improve: 19 of 30 missions with typed order, 15 with exact and 13 with nearest-next. For exact against typed there were 7 against 3 discordant lists, so $p = 2 \times (C(10,0)+C(10,1)+C(10,2)+C(10,3))/2^{10} = 2 \times 176/1024 = 0.34$, not significant. Ordered routes consist of short hops that often start with a turn into an aisle, which is exactly the case that fails the default progress check (Section 5.2).

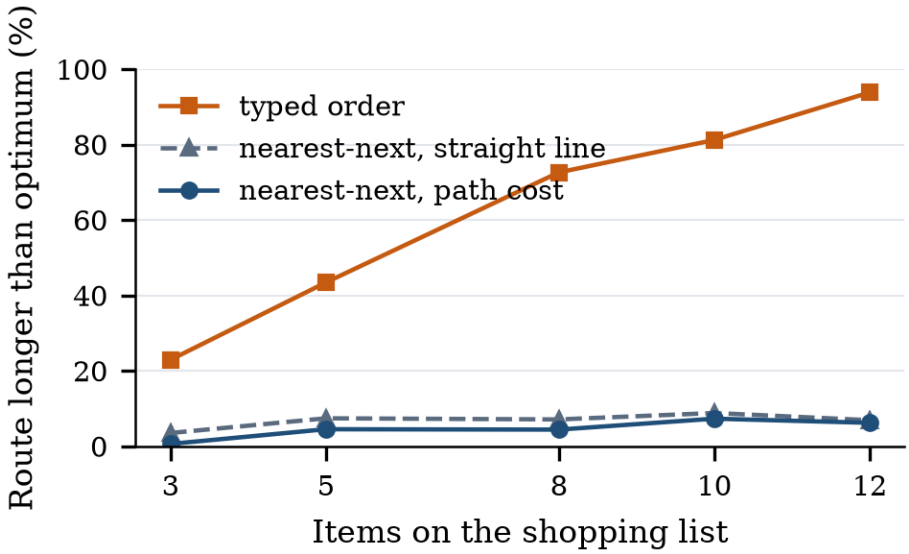


Figure 5.5 : Offline route length relative to the exact optimum (C1, 50 lists per size)

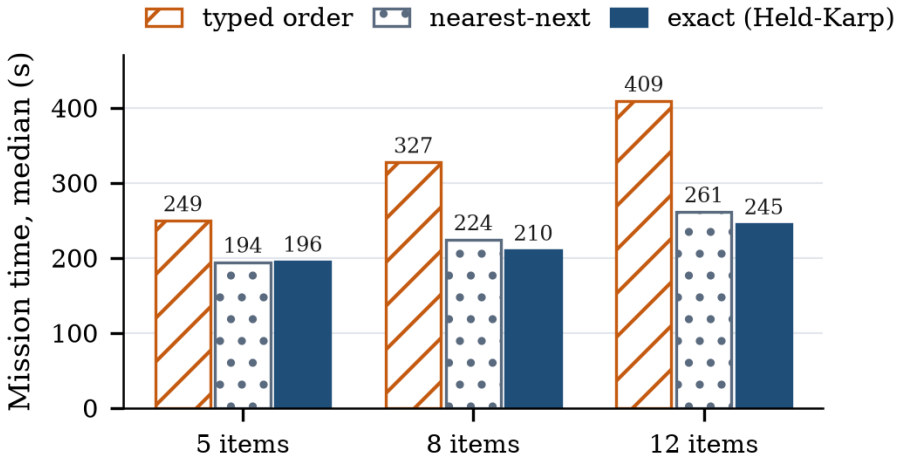


Figure 5.6 : Median mission time by ordering strategy and list size (C2)

5.7 Phase 2: Progress Check and Exact Ordering

Both variants drove the 30 A2 lists (seed 7) on the same stack, changing only a copy of the controller configuration (Table 5.3, Figure 5.7, Figure 5.8). The relaxed check alone removed all 62 aborts and raised first-attempt arrivals from 67% to 96%, but missions took 3.8% longer on the 28 lists both completed ((287.5 − 277.0)/277.0, $p = 1.4 \times 10^{-4}$), and in one leg the robot pressed against a shelf for 42 s. With exact ordering added, A2-final completed all 30 lists and all 144 stops with no contact.

Measure	A2	A2-pc	A2-final
Progress check	0.5 m / 10 s	0.25 m / 20 s	0.25 m / 20 s
Visiting order	Typed	Typed	Exact
Missions completed	29/30	29/30	30/30
Stops reached	143/144	140/141	144/144
Reached at first attempt	96 (67%)	136 (96%)	133 (92%)
Recoveries (aborts)	97 (62)	4 (0)	13 (2)
Median mission time (s)	255	259	218
Mean driven distance (m)	18.6	18.9	11.6
Shelf contacts	0	1	0
Mean $\|\dot{\omega}\|$ (rad/s²)	0.211	0.231	0.273
Steering reversals per m	2.07	2.13	3.01
Loc. error med./max (m)	0.029/0.117	0.025/1.149	0.035/0.140

Table 5.3 : Original stack and phase-2 variants on the 30 A2 lists

Worked significance test. On the 29 lists that both A2 and A2-final completed, the mean time fell from 281.0 s to 207.4 s, a change of (207.4 − 281.0)/281.0 = −26.2%, and the mean distance from 18.6 m to 11.5 m, −38.2%. A2-final was faster on 27 lists. Ranking the 29 absolute time differences gives $W_{pos} = 15$ and $W_{neg} = 420$ (check: $n(n+1)/2 = 435$). From Eq. (4.14), $\mu_W = 29 \times 30/4 = 217.5$ and $\sigma_W = \sqrt{(29 \times 30 \times 59/24)} = 46.2$, so $z = (15 − 217.5)/46.2 = −4.38$. The exact signed-rank distribution gives $p = 5.1 \times 10^{-7}$ (the normal approximation

gives 1.2×10^{-5}); for distance $p = 1.2 \times 10^{-7}$. The cost is smoothness: shorter routes contain more turns per metre, so mean $|\omega|$ rose from 0.211 to 0.273 rad/s² (+29%, $p = 8.1 \times 10^{-4}$).

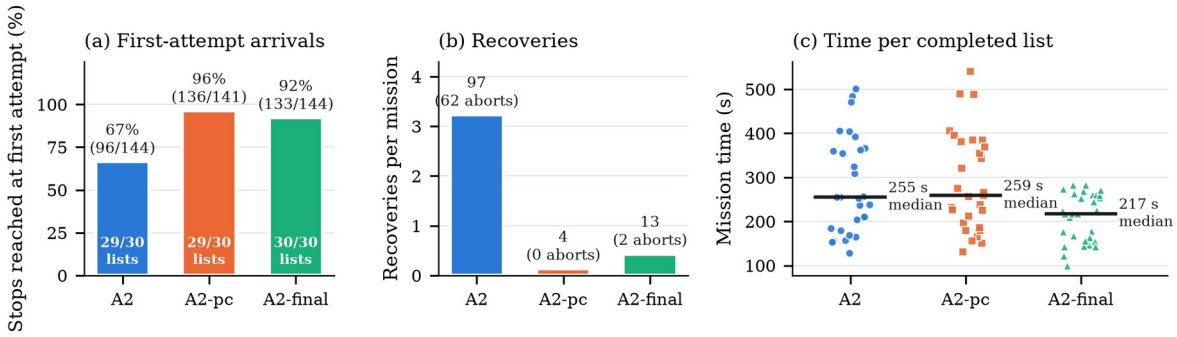


Figure 5.7 : Original stack (A2) and phase-2 variants on the same 30 lists: (a) first-attempt stops, (b) recoveries, (c) mission time

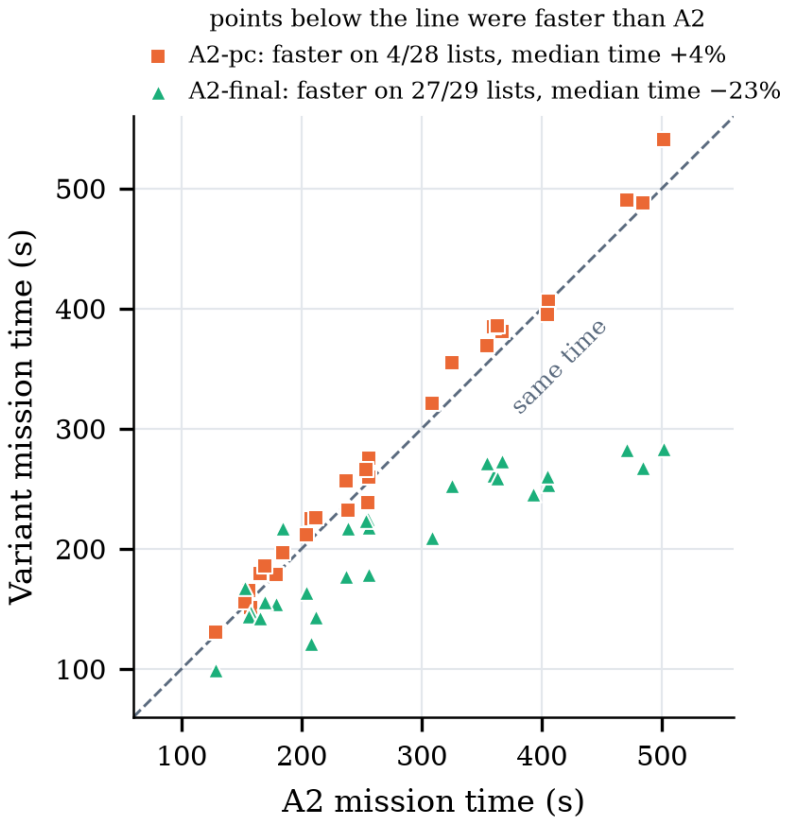


Figure 5.8 : List-by-list mission time of each variant against the original stack

5.8 Bench Hardware-in-the-Loop Test

The bench set-up is shown in Figure 5.9 and a 15-minute session in Figure 5.10; Table 5.4 summarises the measurements. The behaviour tree spent 240 s in LEAD, 29 s in FOLLOW and 616 s in WAIT, with nine starts and seven stops while moving. Over 22 flag changes from the Pi, the tree changed mode a median of 0.12 s after the relay (maximum 0.20 s).

Nine times the Pi cleared both flags; each time the PWM was zero in the first WAIT sample, and in the next 1 s the right encoder counted at most one pulse, i.e. ≤ 1.2 mm of wheel travel by Eq. (4.4) (eight pulses, 9.6 mm, once when stopping from 0.15 m/s).

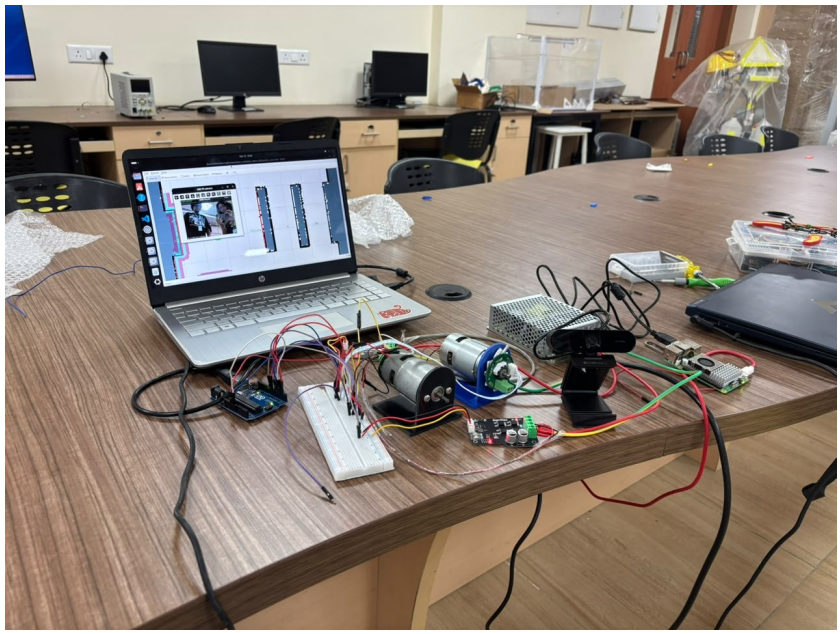


Figure 5.9 : Bench set-up: laptop running the stack, Raspberry Pi 5 with webcam, Arduino Uno, two SmartElex drivers and two encoder motors

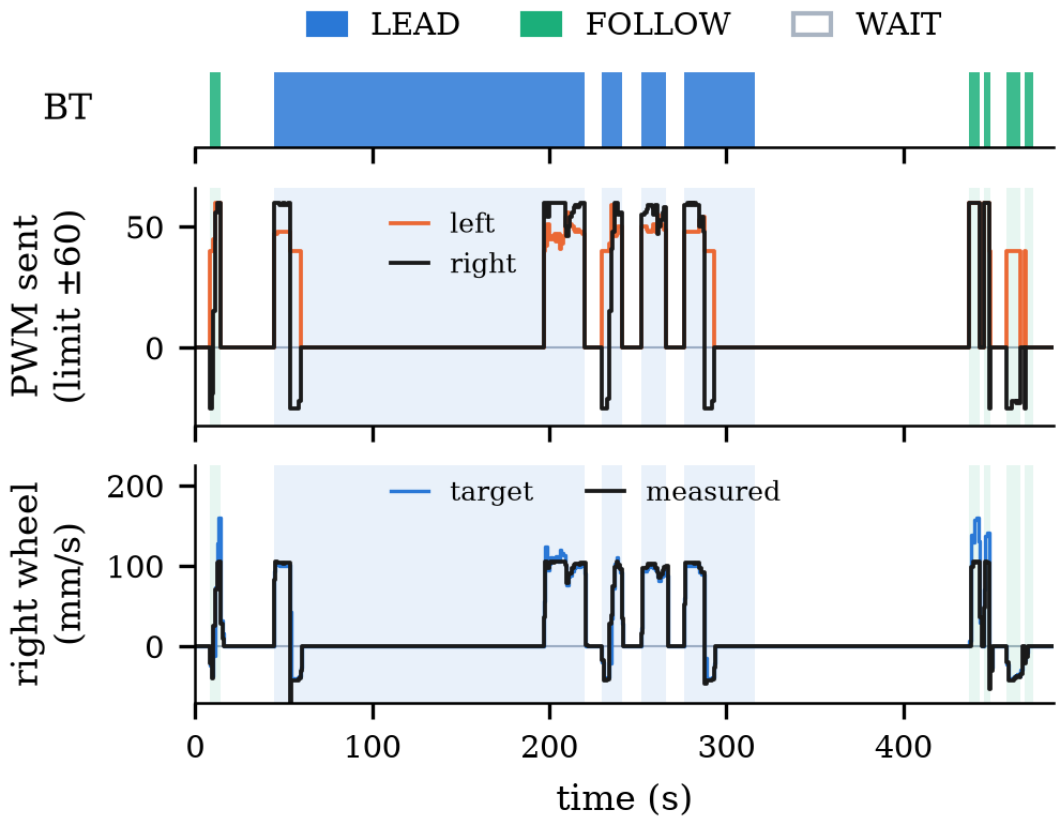


Figure 5.10 : Bench session logged by the motor bridge (1 s samples): behaviour-tree mode, PWM and right-wheel speed

Measurement	Result
Session	15 min; LEAD 240 s, FOLLOW 29 s, WAIT 616 s
Flag → BT mode	Median 0.12 s, max 0.20 s (22 changes)
Stop on WAIT	PWM 0 in first WAIT sample; ≤ 1 pulse (1.2 mm) in next 1 s
Pi data loss	1.5 s silence → relay held WAIT
Right-wheel speed	104 mm/s at 100 mm/s command (97–105)
Left encoder	1 pulse in session (intermittent connection)
Serial link	17 660 commands, 881 reports, 0 errors
Watchdog	No pulses from 1.6 s after commands stopped
Goal-reached race	2 of 4 goals marked complete within 2 ms

Table 5.4 : Bench HIL measurements (16 October 2026)

Speed tracking. At the 0.10 m/s command the right wheel ran at 104 mm/s, a tracking error of (104 − 100)/100 = +4%, about three times the 1.2 mm/s encoder resolution. At 0.15 m/s it was at the ±60 PWM limit and reached about 105 mm/s, so the bench drive cannot yet reach the follower's speed. The left encoder counted one pulse in the session, although it had counted 63–101 pulses/s (75–121 mm/s) in a calibration run 106 minutes earlier, which points to an intermittent connection.

Stopping distance. Taking the software delay as the measured worst case of 0.20 s plus one 0.05 s bridge period, $t_{delay} = 0.25$ s, and an assumed braking deceleration $a_b = 0.5$ m/s² for the loaded cart, Eq. (4.10) gives $d_{stop} = 0.10 \times 0.25 + 0.10^2/(2 \times 0.5) = 0.035$ m at 0.10 m/s and 0.06 m at 0.15 m/s. The camera-to-flag delay was not logged, so each 0.1 s of camera latency adds 10 mm at 0.10 m/s; this delay must be measured before the cart is tested with people. The 1.2 mm bench coast is consistent with the model for unloaded motors.

Pi power. The Pi ran from a 5 V, 2.4 A supply (12 W), 47% of the 25.5 W of the official 5.1 V, 5 A supply [68], which is consistent with the restarts seen when YOLO and OSNet ran together and why OSNet was tested offline. Two shopping lists ran in the simulated store during the session, and the goal-reached race was observed directly: after each arrival the planner replanned to the reached goal every 2 s and the hand-over re-published 'goal reached', so 2 of the 4 goals the shopping manager published were marked complete within 2 ms and skipped.

5.9 Shopping Interface and ROS Link

The interface (Figure 5.11) now drives the stack through arc_ui_bridge. The arrival guard (Eq. (4.15)) directly targets the race of Section 5.8. Nine offline tests passed: product-mapping completeness (60 products → 22 store items), merging, ordering (exact cost ≤ typed cost), stale and distant arrivals ignored, a full trip, pause/stop/resume/home, skip, double start and the server API against stand-in ROS objects. An end-to-end browser test against a simulated robot completed a 5-product list in 4 stops, ignored every stale arrival and raised no script errors. The bridge has not yet been run against the real stack; that test is planned before the review.

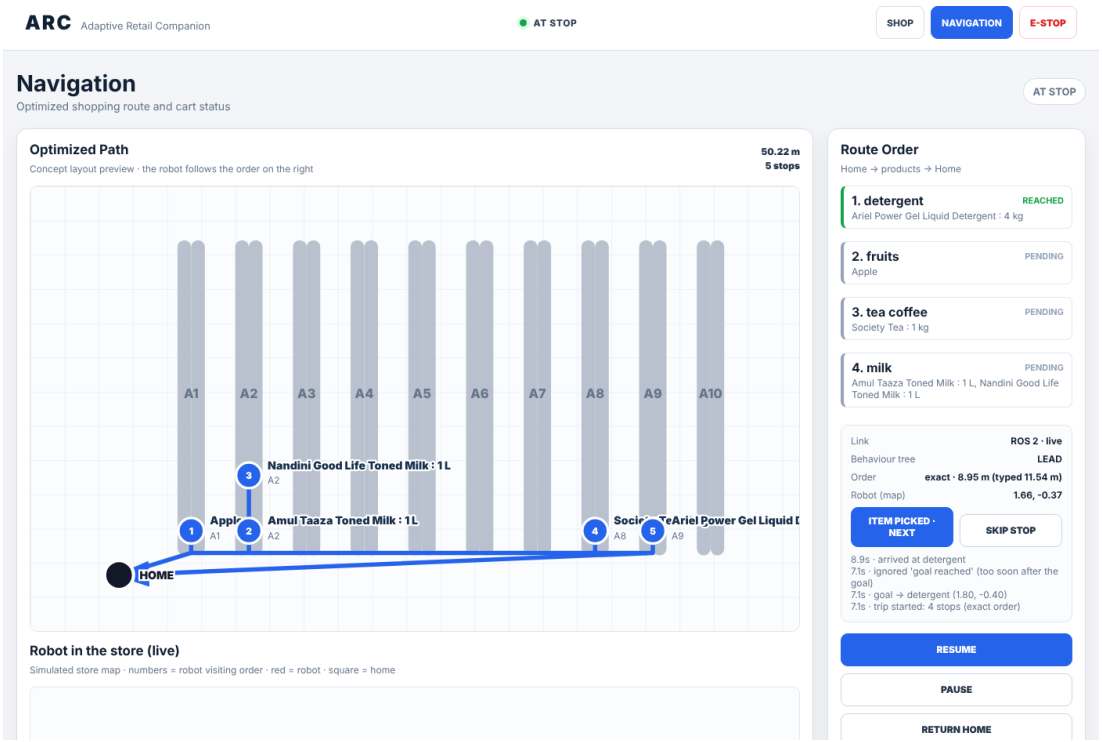
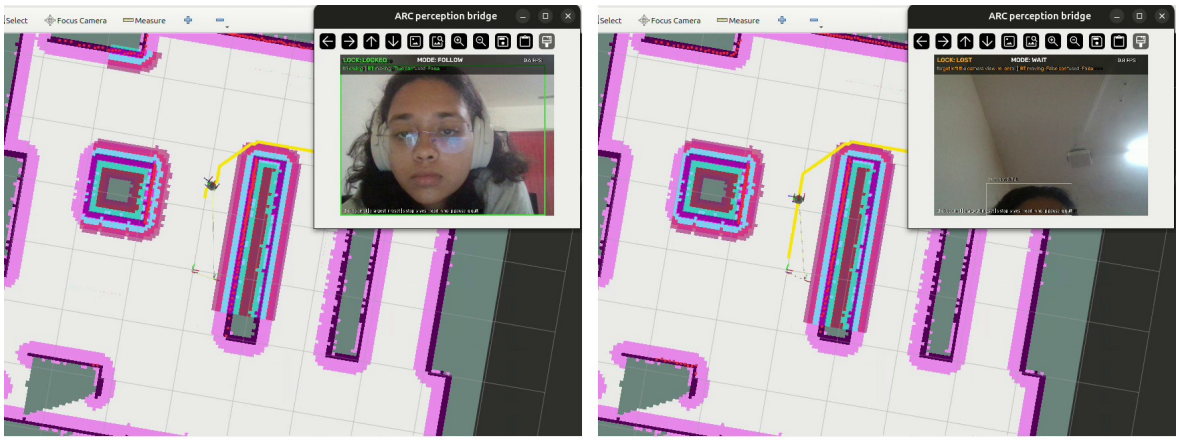


Figure 5.11 : Shopping interface linked to the robot: live route order, behaviour-tree mode and event log (browser test against a simulated robot)

5.10 Person Detection and Target Lock

YOLOv8n and ByteTrack ran live on the Pi in the bench test. In the offline OSNet test the enrolled person changed ByteTrack ID from 1 to 3 after leaving and returning, and the original height-ratio gate (maximum 2.0) rejected the returning target at ratios of 2.56–2.67 despite high appearance similarity. With the gate relaxed to 3.0 the target was re-locked at about 7.7 s, and the lock state was LOCKED for 429 of 465 frames (92.3%). With one person only, this is a diagnostic result, not a re-identification accuracy.

The perception bridge was also run with the laptop webcam against the simulated robot (Figure 5.12). While the target was locked, the bridge reported FOLLOW and the behaviour tree reported the robot as moving; when the target left the camera view, the lock was lost, the mode changed to WAIT and the robot was reported as stopped. This checks the perception-to-behaviour link in the loop with navigation; it gives no timing for the Pi.



(a) (b)

Figure 5.12 : Perception bridge driving the simulated robot: (a) target locked, mode FOLLOW; (b) target left the camera view, lock lost, mode WAIT

5.11 Mechanical Design and Structural Check

5.11.1 Cart geometry, wheel loads and stability

Figure 5.13 shows the measured plan of the retrofitted cart. The modules widen it from the 440 mm body to 667 mm, because each motor projects outboard of its sleeve. A drive module has no interference between any of its parts (adaptor, sleeve, slider, springs, bracket, motor and wheel); the only contact is the slider plate seated flush on the sleeve base. One modelling error was found: the drive wheels reach 36 mm lower than the rear wheels (wheel bottoms at 237.9 mm and 273.9 mm in the model), so as drawn the cart would stand on its drive wheels and pitch back by atan(36/680) = 3.0° until the rear wheels touched. The module height or the rear wheel mounts must be corrected by 36 mm.

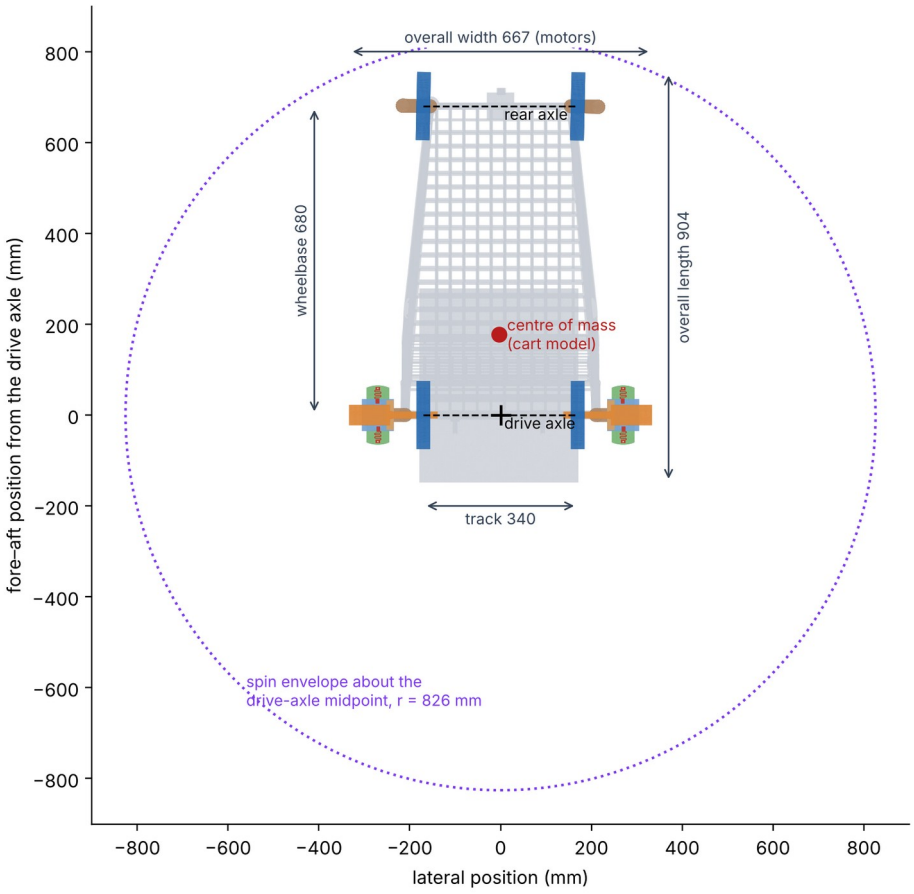


Figure 5.13 : Plan view of the retrofitted cart from the CAD assembly, with the measured dimensions, the centre of mass of the cart model and the spin envelope about the drive axle

The centre of mass of the cart model lies 178 mm behind the drive axle and 502 mm ahead of the rear axle, 600 mm above the floor. Taking a loaded mass of m = 50 kg at that point, Eq. (4.20) gives 73.8% of the weight on the drive axle: $F_v = 50 \times 9.81 \times 0.738/2 = 181$ N on each drive wheel and 64 N on each rear wheel (Table 5.5). The narrow 340 mm track sets the stability limit: the cart tips sideways at a slope of atan(170/600) = 15.8°, or at a lateral acceleration of $9.81 \times 170/600 = 2.78$ m/s². At 0.15 m/s on the tightest 0.2 m turn radius the lateral acceleration is only $0.15^2/0.2 = 0.11$ m/s², a margin of about 25. The 50 kg mass and its position are assumptions; a high or off-centre load in the basket reduces these margins.

Quantity	Calculation	Result
Load on each drive wheel	50 × 9.81 × (502/680)/2	181 N
Load on each rear wheel	50 × 9.81 × (178/680)/2	64 N
Sideways tipping slope	atan(170/600)	15.8°
Tipping slope over the drive axle	atan(178/600)	16.5°
Tipping slope over the rear axle	atan(502/600)	39.9°
Lateral acceleration to tip	9.81 × 170/600	2.78 m/s²
Braking deceleration to tip forward	9.81 × 178/600	2.91 m/s²

Table 5.5 : Static wheel loads and stability of the loaded cart (m = 50 kg at the centre of mass of the cart model)

Braking would tip the cart forward only above 2.91 m/s², nearly six times the 0.5 m/s² assumed for the stopping distance. The traction available at each drive wheel, $0.5 \times 181 = 90$ N for a friction coefficient of 0.5, is 3.9 times the 23.4 N needed on a 1:20 ramp.

The cart's real footprint was re-run through the store-map footprint check (Table 5.6). With the measured width, driving straight needs a 0.384 m clearance radius; all 9 stops remain reachable and routes are 9.6% longer than on the test grid. A differential-drive cart turns about the midpoint of its drive axle, and from Eq. (4.21) its rear corners sweep $r = \sqrt{(755^2 + 334^2)} = 826$ mm, a 1.65 m circle. With that radius only 3 of the 9 stops can be reached, so the cart must manoeuvre in the aisles rather than spin on the spot.

Footprint	Radius (m)	Stops reachable	Goal moved (m)	Route longer
TurtleBot stand-in (planner)	0.200	9/9	0	1.5%
Cart, driving straight	0.384	9/9	0	9.6%
Cart, turning about its centre	0.562	9/9	0.85	5.5%
Cart, spin about the drive axle	0.826	3/9	0.74	—

Table 5.6 : Store-map footprint check with the measured cart dimensions

5.11.2 Drive sizing

With the 150 mm wheels, the LEAD and follow speeds of 0.10 and 0.15 m/s need $60v/(\pi D) = 12.7$ and 19.1 rpm at the wheel. Using Eq. (4.11) with m = 50 kg, $C_{rr} = 0.02$ and a = 0.25 m/s², the tractive force is $F = 50(0.25 + 9.81 \times 0.02) = 22.3$ N on a flat floor and 46.8 N on a 1:20 ramp ($\theta = 2.86°$), shared by the two drive wheels. With $r_w = 0.075$ m the torque per wheel is 0.84 N·m on the flat and 1.75 N·m on the ramp, and the power per wheel at 0.15 m/s is 3.5 W. These are the requirements against which the motor's rated torque and speed must be checked; the bench motors drove smaller 80 mm wheels.

5.11.3 Mount fit and spring latch

Without the motor and wheel, one mount weighs 0.49 kg in aluminium or 1.42 kg in steel (Table 4.5); the sleeve is 72% of it. The guide fit from Eq. (4.18) is loose: the slots are 15.2 mm wide and the rails 13.0 mm, so $c_s = (15.2 - 13.0)/2 = 1.1$ mm per side, and the slot floor (radius 19.5 mm) sits 1.0 mm inside the rails (radius 20.5 mm). At the mean rail radius $r_m \approx 25$ mm this lets the module rotate by $\theta_p \approx 1.1/25 = 0.044$ rad, about 2.5°, about the vertical axis. For a driven wheel that is a steering error, so the clearance should be reduced once the

manufacturing tolerances are known; 0.2 mm per side would limit the rotation to about 0.5°.

For an illustrative stainless-steel strip (E = 193 GPa, w = 10 mm, t = 0.5 mm, L_s = 30 mm) deflected by 1.5 mm, Eq. (4.19) gives $F = 193\,000 \times 10 \times 0.5^3 \times 1.5/(4 \times 30^3) = 3.35$ N per spring and $\sigma = 3 \times 193\,000 \times 0.5 \times 1.5/(2 \times 30^2) = 241$ MPa. These dimensions are examples, not taken from the spring model. The stress exceeds the yield strength of annealed austenitic stainless steel, so a spring-temper grade is needed. Two springs give about 6.7 N, less than the weight of a module with its motor and wheel when the cart is lifted, so the latch must hold by a positive catch that the springs keep engaged, not by spring force or friction.

5.11.4 Finite-element validation

SimScale reported a maximum von Mises stress of 14.6 MPa (Figure 5.14). That value is a single node at the corner of the fixed support; the next five highest values, 6.6–10.2 MPa, are also at that constraint corner, where a rigid support produces an artificial stress singularity. Away from the support the peak is 4.25 MPa, at the plate-root fillet, the stress at the plate root is typically 2.4–2.8 MPa (Figure 5.15), and the maximum displacement is 0.020 mm, at the free end of the bracket.

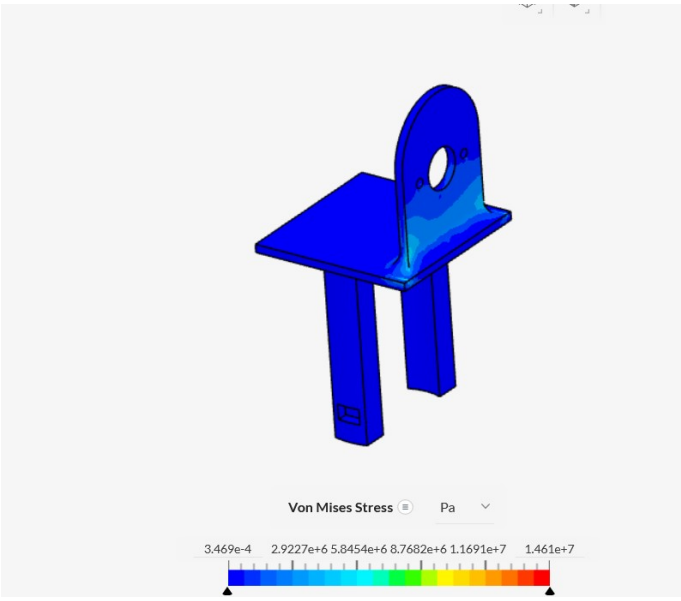


Figure 5.14 : SimScale von Mises stress in the slider and wheel bracket (legend in Pa)

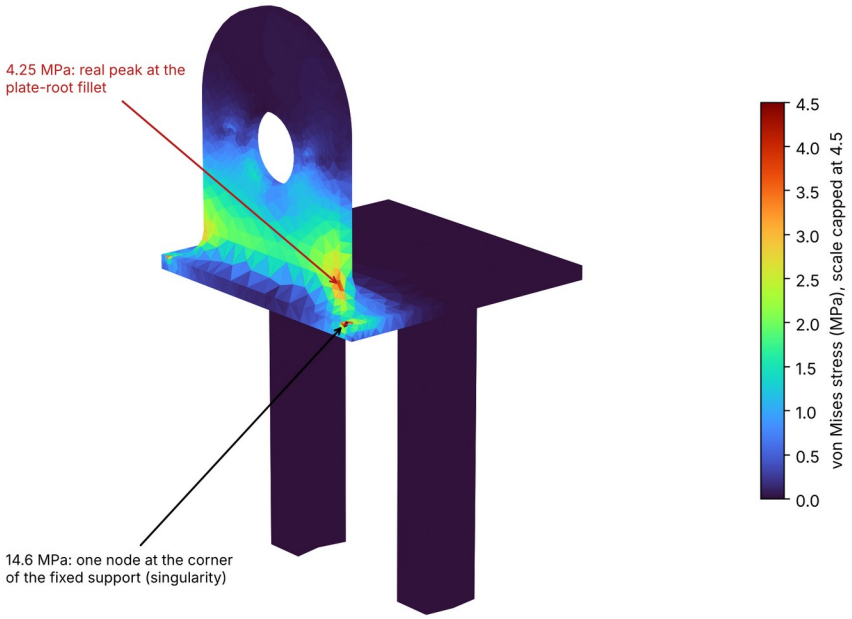


Figure 5.15 : The SimScale result re-plotted with the colour scale capped at 4.5 MPa, showing the peak at the root fillet and the support singularity

Beam theory (Eq. (4.16)) and the independent scikit-fem model reproduce this result (Table 5.7). At the end of the root fillet, 25 mm from the load, $\sigma = 10 \times 25/102.1 = 2.45$ MPa. With the load at $a = 32.5$ mm and the free end at $L = 65$ mm, $\delta = 10 \times 32.5^2 \times (3 \times 65 - 32.5)/(6 \times 70\,000 \times 178.6) = 0.023$ mm; the beam model ignores the stiffening of the root fillets and so slightly over-predicts. The scikit-fem model gives 0.017 mm and a root stress of 2.1–2.5 MPa. The three methods agree on the stress at the root to within about 15%; the higher SimScale peak at the fillet corner depends on how the 10 N load was distributed over the hole, which is not stored in the exported results.

Quantity	SimScale	scikit-fem	Beam theory
Root stress	2.4–2.8 MPa	2.1–2.5 MPa	2.45 MPa
Free-end deflection	0.020 mm	0.017 mm	0.023 mm
Peak at root fillet	4.25 MPa	2.5 MPa	—
Reported maximum	14.6 MPa (singularity)	—	—

Table 5.7 : Bracket under the SimScale load (10 N normal to the plate and 10 N in its plane)

5.11.5 Bracket under the design wheel load

In service the bracket carries the wheel load $F_v = 181$ N at the measured 71.7 mm offset, a moment of $M = 181 \times 71.7 = 12\,978$ N·mm, 40 to 50 times the moment in the SimScale case. Beam theory gives a nominal stress of 12 978/102.1 = 127 MPa along the whole plate. The scikit-fem model under this load (Figure 5.16, Table 5.8) gives 139 MPa at the root fillet and 246 MPa in the narrow ligament between the motor hole and a motor-screw hole, where the holes remove more than half of the 50 mm plate width. Refining the mesh from 96 231 to 167 157 degrees of freedom changed the peak by 0.1% and the root stress by 2.6%, so the result is mesh-converged. The plate rotates by 1.59° under the motor in aluminium (beam theory: 1.93°), which tilts the wheel by the same angle, and the free end deflects by 1.5 mm.

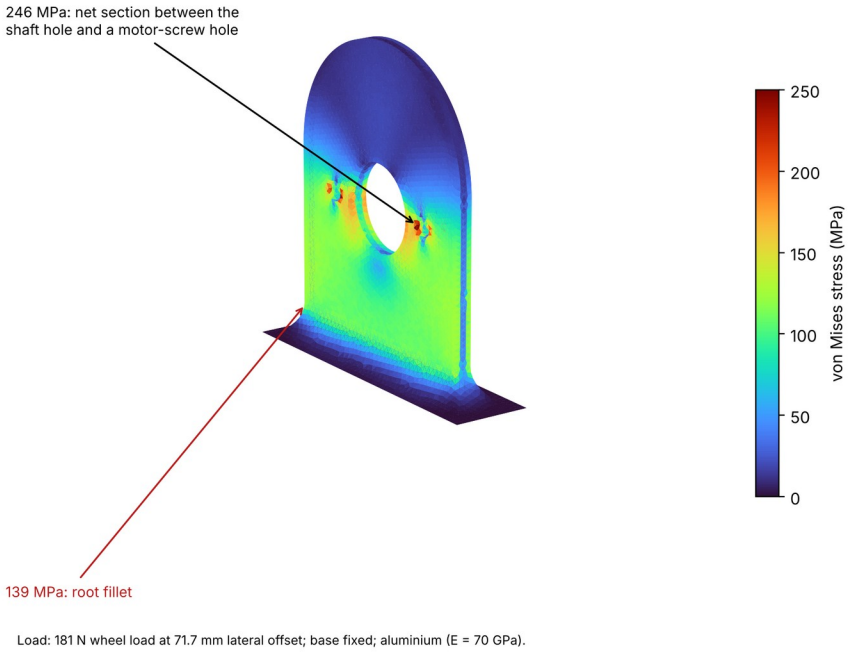


Figure 5.16 : Independent finite-element model of the wheel bracket under the design wheel load (181 N at 71.7 mm offset)

Quantity	Result	SF Al 6061-T6	SF steel E250
Nominal bending stress (beam)	127 MPa	2.2	2.0
Root fillet (FEA)	139 MPa	2.0	1.8
Ligament at motor holes (FEA)	246 MPa	1.1	1.0
With bump factor 2	493 MPa	0.6	0.5
Wheel tilt, Al / steel	1.59° / 0.56°	—	—

Table 5.8 : Wheel bracket under the design wheel load (scikit-fem; stresses are the same in aluminium and steel)

Against the 276 MPa yield strength of 6061-T6 aluminium [66] and 250 MPa for E250 steel [62], the bracket as drawn has a safety factor of only about 1 at the motor holes, would yield under a bump load, and is far above the 96.5 MPa fatigue strength of the aluminium. The 3.5 mm bracket is therefore not adequate for the measured wheel offset. The stress scales with the moment and with $1/t^2$, so the options compare as follows (same hole pattern, factors from the FEA): a 6 mm plate lowers the peak to 84 MPa (safety factor 3.3 in

aluminium); moving the wheel to 20 mm from the plate lowers it to 69 MPa (4.0); both together give 23 MPa (11.8); and a fork that supports the shaft on both sides of the wheel removes the moment, leaving an in-plane stress of about 181/(23.6 × 3.5) = 2.2 MPa. A 6 mm steel plate with the wheel closer to it is the simplest change.

5.11.6 Motor attachment and shaft

Two further points follow from the measured geometry. First, the two motor screws sit fore and aft of the motor hole, on the axis about which the wheel-load moment acts, so they have no lever arm to resist it; the moment is carried only by the motor flange bearing on the plate and by the motor boss in its hole. The screw pattern should be turned through 90°, or four screws used. Second, the 15 mm shaft carries the wheel 69.95 mm from the bracket face, so from Eq. (4.22) its bending stress is $\sigma_s = 32 \times 181 \times 69.95/(\pi \times 15^3) = 38$ MPa, acceptable for a steel shaft, but the 181 N load at that overhang must be within the gearbox's permissible overhung load, which has to be confirmed from the motor datasheet.

5.12 Integration-level Findings

Five integration-level properties mattered more than the choice of any one algorithm: (1) Nav2's default progress check, never overridden by ARC, caused all 62 aborts; (2) the hand-over has no replanning or recovery; (3) a goal-reached race lets a stale arrival skip goals; (4) lidar odometry does not recover from a failed match; and (5) lidar odometry protects re-localisation. The recommendation is to adopt both phase-2 changes together, a progress check of 0.25 m in 20 s and exact path-cost ordering, paired with contact or slip detection on the cart [64].

5.13 Innovation and Technical Depth

ARC's originality lies in the system and the evidence rather than in a new algorithm. The project combined a retrofit concept, a full ROS 2 stack, exact route ordering, on-board person detection, a bench drive and a web interface, and evaluated them with a baseline-first protocol in which every fix was tested on the same tasks and seeds as the original. Faults were traced to specific code and parameters and explained by engineering models: the turn-time calculation predicts the progress-check aborts, the Held–Karp complexity shows the optimum is affordable on board, and the encoder model converts pulses into the

measured stop and speed figures. Integrity checks in the test chain (verified obstacle removal, goal-matched arrivals, automatic re-checks) were themselves a technical contribution. The mechanical design was checked in the same way: the CAD assembly was measured in code, and an independent finite-element model, validated against SimScale and beam theory, showed before any part was made that the wheel bracket must be stiffened.

5.14 Cost-Benefit and Sustainability Impact

The measured benefit of the recommended fix is a 26% shorter and 38% less-driven shopping trip with no lost missions. Drive energy scales with distance: at the assumed 50 kg and $C_{rr} = 0.02$, rolling energy per trip falls from $m\ g\ C_{rr}\ d = 50 \times 9.81 \times 0.02 \times 18.6 = 182$ J to 114 J, and any constant electronics load is on for 26% less time. The fix costs only configuration and software. ARC's bill of materials is not final, so a price comparison with commercial smart carts, which are quote-priced [67], is not yet possible (Table 5.9).

Option	Benefit	Cost or risk
ARC retrofit kit	Reuses existing carts; module serviced separately	Mount must fit different carts; BOM not final
Phase-2 software fix	−26% time, −38% distance, 30/30 lists	None in hardware; less smooth turning
Commercial smart cart	Item recognition and checkout	Quote-priced; no autonomous assistance
Simulation-first testing	Repeatable, low hardware dependence	Cannot model the real cart, floor and people

Table 5.9 : Preliminary cost-benefit assessment

5.15 Environmental and Societal Impact

Retrofitting extends the life of carts a store already owns, and shorter routes reduce energy per trip. For users, a cart that leads, follows and waits reduces the effort of shopping and supports independent access to stores. These societal benefits are intended outcomes; they

require trials with older users, privacy-aware perception and certification against ISO 13482 before they can be claimed.

CHAPTER 6

CONCLUSION

- ARC's navigation stack localised to 2–3 cm and completed 29 of 30 shopping missions in simulation, but needed 97 recoveries; a kinematic calculation showed that Nav2's default progress check cannot be met after a 180° in-place turn at 0.10 m/s.
- Exact path-cost ordering cut driving time by 25–41%. Combined with a progress check of 0.25 m in 20 s it completed all 30 lists in 26% less time and 38% less distance ($p = 5.1 \times 10^{-7}$), meeting the evaluation objectives.
- On the bench, Pi-based person detection drove the behaviour tree and real motors; the wheels stopped within one 1 s telemetry period of every WAIT, and the shopping interface was linked to the stack through a new ROS 2 bridge.
- A quick-release drive mount was designed in CAD; a structural check showed that the loaded cart is stable but that the 3.5 mm wheel bracket must be thickened, or the wheel moved closer to it, before fabrication.

Scalability. The kit is software-defined: the same stack, test kit and interface apply to any store whose map can be built, and the mount can be adapted to other carts or to hospital and warehouse trolleys. Mass production needs a standard adjustable mount, a costed BOM, an independent motor-power E-stop and certification.

Technical hurdles. First, a goal-reached race: the latched 'goal reached' message and the planner's 2 s replan made stale arrivals skip goals; this was traced event by event and the UI bridge now accepts an arrival only near the goal and at least 3 s after it was sent. Second, the left encoder dropped out during the bench session, so that wheel ran without feedback; it was traced to an intermittent connection by comparison with the earlier calibration. Pi restarts under combined YOLO and OSNet load were traced to a 12 W supply.

Future work. Adopt the progress-check and ordering fixes in ARC; add replanning to the hand-over and odometry recovery after contact; restore left-wheel feedback; measure the camera-to-stop delay and the Pi frame rate; mount a 2D lidar on the cart; run the UI bridge on the real stack; and test two-person re-identification and trials with older users.

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APPENDIX A

This appendix lists the parameters used in the calculations of Chapters 4 and 5 and the mapping from interface products to store items used by the UI bridge.

A.1 Parameters Used in the Calculations

Parameter	Value	Source
Wheel diameter D	0.08 m	M (bench)
Track width B	0.16 m	M (bench)
Pulses per wheel revolution P_w	7 × 30 = 210	M (motor, gearbox)
Telemetry period Δt	1 s	M (firmware)
LEAD / follow speed	0.10 / 0.15 m/s	M (configuration)
ω_{max}, α_{max}	0.5 rad/s, 1.0 rad/s²	M (configuration)
Progress check r, T	0.5 m, 10 s → 0.25 m, 20 s	M (Nav2 default, phase 2)
Software stop delay	≤ 0.20 s + 0.05 s	M (logs, bridge rate)
Braking deceleration a_b	0.5 m/s²	A
Mass m	50 kg	A
Rolling resistance C_{rr}	0.02	A
Acceleration a	0.25 m/s²	A
Ramp	1:20 (2.86°)	A
Gearbox efficiency	80%	A
Bracket plate b × t	50 × 3.5 mm	M (CAD)
Root to shaft L_h	32.5 mm	M (CAD)
Guide slot / rail width	15.2 / 13.0 mm	M (CAD)
Wheel offset e	71.7 mm	M (CAD)
Wheelbase / drive track	680 / 340 mm	M (CAD)
Centre-of-mass height	600 mm	M (CAD cart model)
Loaded mass m	50 kg	A

Parameter	Value	Source
Wheel load F_v	181 N	M (CAD) with m = 50 kg
Wheel radius r_w	75 mm	M (CAD)
Friction coefficient	0.5	A
E aluminium / steel	70 / 200 GPa	M (FEA) / A

Table A.1 : Parameters used in the calculations (M = measured or configured, A = assumed)

A.2 Interface Product to Store Item Mapping

Product IDs	Store item
FP001–003 / FP004–006	fruits / vegetables
DY001–004 / DY005–006	milk / curd_paneer
BK001 / BK002–005 / BK006	bakery / biscuits / breakfast_cereal
GR001–004, GR006 / GR005	atta_flour / rice
GS001–003 / GS004 / GS005–006	dal_pulses / cooking_oil / sauces
PF001 / PF002 / PF003, 004, 006 / PF005	pasta / biscuits / jam_spreads / bakery
SN001–006	snacks
BV001, 002, 004 / BV003 / BV005–006	tea_coffee / juice / water_beverages
HC001–003 / HC004–006	detergent / cleaning
PC001, 003, 004, 006 / PC002, 005	toiletries / soap_shampoo

Table A.2 : Mapping of the 60 interface products to store items